

Received 6 January 2026, accepted 21 January 2026, date of publication 28 January 2026, date of current version 11 February 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3658687

RESEARCH ARTICLE

Multi-Task FT-Transformer: A High-Performance and Interpretable Framework for Traffic Accident Severity Prediction

YAHUI WANG¹, ZHOUSHUO LIANG, AND YUE HE, (Member, IEEE)

School of Medical Technology, Beijing Institute of Technology, Beijing 100081, China

Corresponding author: Yahui Wang (yhwangmn@bit.edu.cn)

This work was supported by the Foundation of Beijing Natural Science Foundation under Grant 9244037, in part by Beijing Institute of Technology Research Fund Program for Young Scholars under Grant 3320012222316, and in part by the Foundation of the National Key Laboratory of Human Factors Engineering under Grant HFNKL2023WW02.

ABSTRACT Traffic accidents are a significant concern worldwide, leading to severe injuries, fatalities, and property damage. Predicting the severity of these outcomes is essential for enhancing road safety. However, the complex interdependencies among injuries, fatalities, and property damage, coupled with the influence of various dynamic factors, pose challenges for prediction. To address this issue, This study proposes a multi-task feature token transformer model with uncertainty-weighted loss optimization to predict the severity of injuries, fatalities, and property damage in traffic accidents. The model is trained on a large-scale dataset comprising urban road accidents in China, incorporating critical factors such as driver characteristics, vehicle types, road conditions, and environmental influences. The multi-task learning framework enhances information sharing across the prediction tasks for fatalities, injuries, and property damage, resulting in improved accuracy compared to single-task frameworks. Furthermore, we apply SHapley Additive Explanations to quantify feature importance and analyze how different input factors influence accident severity predictions, enhancing model transparency and interpretability by providing a clear understanding of the decision-making process. The results show that the proposed multi-task FT-Transformer model achieves good accuracy in all severity prediction tasks, with 70.31% for injury severity, 90.38% for fatal accident severity, and 80.79% for property damage. This framework helps policymakers target high-risk areas and optimize resource allocation, improving road safety and reducing traffic incidents.

INDEX TERMS Traffic accident severity prediction, multi-task learning, deep learning, FT-Transformer, explainable artificial intelligence.

I. INTRODUCTION

Traffic accidents remain one of the leading causes of bodily harm, death, long-term disability, and property loss, imposing persistent burdens on both the economy and healthcare systems [1]. According to the China Statistical Yearbook, in 2020 there were 244,674 road traffic accidents nationwide, resulting in 61,703 deaths and 250,723 injuries; by 2022, the total number of crashes had increased to 256,409, with direct property losses of about 1.24 billion yuan [2], [3]. These data highlight the continuing severity of traffic

accidents in China and emphasize the urgent need for more effective traffic safety measures. To mitigate these impacts, it is crucial to understand the factors influencing accident severity, such as driver behavior, vehicle type, and environmental conditions. These factors interact in intricate ways, collectively shaping the outcomes of accidents. By identifying and analyzing these patterns, it is possible to predict and potentially reduce the severity of accidents [4]. Consequently, accurately predicting accident severity is key to intelligent traffic safety management. Focusing on severity rather than just occurrence helps identify high-risk scenarios and enables proactive safety measures. Thus, timely and accurate severity prediction is essential.

The associate editor coordinating the review of this manuscript and approving it for publication was Yanli Xu².

Previous studies have applied statistical and machine learning methods to predict injury severity in traffic accidents, linking it with driver, road, vehicle, and environmental factors [5], [6]. Understanding these relationships facilitates the creation of more precise models for predicting accident severity. However, several challenges persist in traffic accident severity prediction. Whereas some research primarily focuses on predicting injury and fatal accident severity, studies addressing the severity of property loss are relatively less common [7]. Additionally, factors affecting the severity of property loss have received comparatively less attention in previous research on traffic accidents. Furthermore, recent studies have focused on single severity indicators, such as injury or property damage, whereas the potential interrelationships between injury severity, fatal accident severity, and property loss, as well as the interpretability of influences under different severity levels of injury, fatality, and property damage, have not been extensively explored. Multi-task learning can capture these relationships by simultaneously addressing multiple related tasks, which can improve prediction accuracy and offer a more comprehensive understanding of the factors influencing traffic accidents [8], [9]. In parallel, we conduct interpretability analysis to uncover how different factors contribute to injury severity, fatal accident severity, and property damage. Understanding these relationships helps in identifying critical risk factors and improving model transparency.

Since the severity of injuries, fatalities, and property loss in traffic accidents are often interconnected, a multi-task model can capture these relationships more effectively than single-task models [10], [11]. Moreover, multi-task learning effectively utilizes limited data by sharing information across related tasks, improving predictions for rare events like severe property damage. However, whereas multi-task learning has been applied to traffic accident severity prediction, challenges remain in optimizing tasks such as injury, fatality, and property damage levels, as well as improving the interpretability of different task predictions. In this study, an uncertainty-weighted loss method is utilized to achieve an optimal balance between tasks by using uncertainty-weighted loss optimization, which adjusts task weights based on their inherent uncertainty to improve overall performance. Furthermore, understanding a prediction model's explainability is crucial for practitioners to identify relevant factors [12]. Therefore, this study uses SHAP (SHapley Additive exPlanations) at the sub-class level, focusing on severity categories such as fatalities, injuries, and property damage, to analyze how factors contribute to accident severity, providing insights into the key factors influencing traffic accident severity and aiding informed decision-making.

In this research, we introduce a multi-task feature token transformer (FT-Transformer) framework designed to predict the severity of injuries, fatalities, and property damage resulting from traffic accidents. In contrast to existing single-task or conventional multi-task models, our proposed FT-Transformer leverages an uncertainty-weighted loss

function to dynamically balance task-specific weights, thereby enhancing information sharing across tasks. This approach is particularly effective in multi-task environments where severity categories, such as fatalities and property damage, exhibit inherent interdependencies. The uncertainty-weighted mechanism optimizes model performance across all severity categories, improving both predictive accuracy and generalizability, especially in rare, high-severity cases. To identify the factors most significantly influencing the three types of severity, we employed SHAP (SHapley Additive exPlanations). Experimental results on Traffic accident data from China reveal that our model delivers superior prediction accuracy and interpretability, consistently surpassing baseline models.

In this study, we model accident severity along three outputs (injury severity, fatal accident severity, and property-loss severity) using a unified multi-task framework. In official reporting, crashes with bodily injury or death are classified as casualty accidents, whereas crashes with property damage only are recorded as material-damage accidents. Post-accident medical or rehabilitation expenditures are not available in our police-reported dataset and are therefore beyond our analysis. This study differs from prior work by formulating accident severity prediction as a unified multi-output learning problem and by introducing an uncertainty-weighted optimization strategy that dynamically balances the three severity outcomes. This design enables the model to explicitly capture the statistical interdependencies among injury, fatality, and property-loss severities, providing an application-oriented extension of existing single-task and fixed-weight multi-task approaches for traffic accident severity prediction.

Specifically, our contributions are as follows:

1. We develop an application-oriented multi-task FT-Transformer framework for predicting traffic accident severity across injuries, fatalities, and property damage. This model leverages an uncertainty-weighted loss function to balance task-specific contributions dynamically, enhancing prediction accuracy and generalizability.
2. We apply SHAP at the sub-class level to identify how factors such as driver characteristics, environmental conditions, vehicle type, and road structure influence accident severity, enabling targeted safety interventions for each severity level.

The structure of the paper is as follows. Section II provides a review of the literature on factors affecting traffic accidents, models predicting accident severity, and the interpretability of these predictions. Section III describes the traffic accident dataset employed in this research, outlines the proposed multi-task deep learning architecture, explains the multi-task loss optimization method, and presents the interpretability framework. Section IV showcases the findings from quantitative experiments regarding prediction quality and includes case studies on the interpretability of the proposed model. Finally, Section V concludes the paper and explores possible avenues for future research.

II. RELATED WORKS

A. CONTRIBUTING FACTORS IN TRAFFIC ACCIDENTS

Traffic accident severity results from the interplay of various factors, such as environmental conditions, driver behavior, vehicle characteristics, and road features. Understanding these multi-dimensional interactions is fundamental for modeling accident causality and predicting severity outcomes [13].

Driver behavior is a critical determinant of traffic accident severity. Impaired driving, such as alcohol consumption and fatigue, significantly increases accident risk. These behaviors impair visual attention, slow perception and reaction time, and reduce hazard anticipation, thereby elevating the probability of severe crashes. For instance, Liang et al. found a clear link between these behaviors and higher accident rates in Suzhou, Anhui Province, China [14]. Comparable behavioral risks have also been reported in other regions (for example, Iran), which illustrates the universality of human-factor influences rather than implying a direct cross-national comparison [15]. Furthermore, the literature consistently shows that male drivers face a greater risk of sustaining fatal injuries than female drivers [16]. Additionally, inexperienced drivers face a higher risk of severe accidents.

Environmental conditions, including extreme temperatures and humidity, significantly contribute to motor vehicle accidents. High temperatures, low air pressure, and elevated humidity can degrade driver alertness and alter tire-road friction, increasing crash likelihood. Research has shown that high temperatures, low air pressure, and moderate-to-high humidity in China, along with high humidity during winter in Kuwait, greatly increase the likelihood of traffic accidents [17]. Additionally, daylight plays a crucial role in injury severity, with daytime crashes generally resulting in less severe injuries compared to nighttime crashes, mainly due to better visibility and shorter perception and reaction time [18]. Taken together, environmental stressors often act as amplifiers of driver errors and indirectly escalate crash severity.

Road characteristics are significant determinants of traffic accident severity. Complex or poorly designed geometries, such as curves, slopes, and multi-lane intersections, increase driver workload and reduce maneuvering margins, contributing to a heightened risk of serious injuries [19]. Moreover, visual obstructions, lack of traffic controls, and features like uncontrolled crossings, curves, and slopes further heighten injury risk [20]. By contrast, speed enforcement (for example, speed cameras) and controlled intersections tend to mitigate high-severity events [21]. Infrastructural conditions frequently interact with driver behavior, amplifying or attenuating risk depending on local design and regulation.

Vehicle-related factors are critical in influencing the severity of traffic accidents. The type of vehicle involved is a significant determinant, with larger vehicles such as trucks or minibuses generally leading to more severe outcomes, particularly in fatal accidents [22]. Additionally, specific vehicle-related problems like brake failure, rollovers, and tire bursts are key contributors to traffic incidents, because

such mechanical faults often induce loss of control or stability [23]. Moreover, in pedestrian-vehicle collisions, the vehicle's speed, weight, and design affect injury severity, with larger, heavier vehicles posing a greater risk of fatal injuries to pedestrians [24]. It is also worth noting that airbag deployment, while designed to reduce injuries, often serves as an indicator of crash intensity when triggered, reflecting the underlying kinetic energy rather than a purely protective effect.

Affecting traffic accident severity, most focus on injuries and fatalities, with limited research addressing property damage factors. However, accidents frequently involve simultaneous injuries, fatalities, and property losses, and these outcomes are interdependent. Given that traffic accidents often involve simultaneous occurrences of injuries, fatalities, and property damage, analyzing factors linked to all three types of accident severity is crucial. This study advances prior research by using a multi-task classification approach, enabling a more integrated analysis. The proposed framework captures both shared and task-specific patterns across the three outcomes, and through uncertainty-based loss weighting it adaptively balances task contributions to improve prediction accuracy and interpretability by leveraging their interrelationships.

B. TRAFFIC ACCIDENT SEVERITY PREDICTION MODELS

Traffic accident severity models are designed to assess severity levels and identify contributing factors more accurately [25]. These models are crucial for accident prevention, reducing accidents and mitigating injuries, fatalities, and property damage, particularly for vulnerable road users [26]. Recent studies on traffic accident severity prediction predominantly utilize machine learning methods and statistical methods. In addition, deep learning and multi-task learning have been increasingly adopted to address nonlinearity, high-dimensional interactions, and task interdependencies.

Machine learning methods are widely used for injury severity prediction due to their ability to capture nonlinear relationships between input and output data. Lin et al. analyzed 8,859 crashes involving teen drivers on rural West Texas roads using Random Forest and XGBoost models, identifying road category, speed restrictions, and initial harmful incidents as key factors, with XGBoost achieving the highest accuracy [27]. Arhin and Gatiba analyzed 3,307 crashes at unsignalized intersections in Washington, DC, using SVM and Gaussian Naive Bayes, with radial basis SVM achieving the highest accuracy (83.2%) [27]. Chen and Chen applied Logistic Regression, CART, and Random Forest to 9,472 crashes in Taiwan, finding Random Forest most effective, with reverse driving and overtaking as key injury severity factors [28]. However, these machine learning approaches primarily focus on capturing basic nonlinear features and often fail to capture complex spatiotemporal correlations in traffic data. Moreover, their interpretability is limited unless augmented with post hoc explanation tools, which may not fully reveal causal mechanisms.

Deep learning, a rapidly advancing computational approach, is increasingly applied in accident analysis. Widely used models include Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN) [29], [30]. Sameen et al. compared RNN, CNN, and FFNN models for traffic accident severity prediction, with RNN achieving the highest accuracy at 73.76%, outperforming CNN and FFNN [31]. Zheng et al. introduced a CNN model that mapped accident data into a grey image for severity prediction, outperforming models like K-nearest neighbor, logistic regression, and LSTM-RNN using data from Leeds City Council [32]. Deep models can learn hierarchical and temporal representations from high-dimensional inputs, yet they usually require substantial data and careful regularization, and their opaque model nature challenges transparent factor attribution without dedicated interpretability methods.

Multi-task learning (MTL) enhances traffic accident severity prediction by simultaneously addressing related tasks, such as accident occurrence, injury severity, and contributing factors. By sharing information across tasks, MTL improves the model's ability to capture complex relationships, leading to more accurate and reliable predictions. Dong proposed an MTL approach for IoT traffic classification using deep autoencoders to enhance generalization and detect rare attacks better than single-task models [33]. Trirat et al. introduced a multi-task deep learning framework for predicting traffic accident severity, offering accurate predictions by capturing shared and distinct risk factors [34]. Zou et al. developed a spatiotemporal generative inference network using MTL to predict highway traffic speeds, improving long-term prediction accuracy by handling speed heterogeneity and sharing parameters across tasks [35]. These multi-task learning models offer substantial improvements over traditional methods, particularly in capturing complex spatiotemporal correlations and enhancing prediction accuracy across related tasks. Nevertheless, MTL introduces additional complexity in optimization and demands appropriate loss balancing to avoid task domination or negative transfer.

In parallel with these model-level advances, practical deployment of traffic accident severity prediction increasingly relies on Internet of Vehicles (IoV) and vehicular edge computing (VEC) infrastructures. Such systems provide the low-latency communication and distributed computation necessary for real-time safety assessment. Recent studies in the IoV domain demonstrate that distributed deep multi-agent reinforcement learning can enhance cooperative edge caching efficiency [36], while incentive-driven partial offloading and resource allocation schemes further improve the reliability of vehicular data processing [37]. Although these works address system-level communication and resource management rather than severity modeling itself, they offer the computational foundations that enable interpretable and efficient severity prediction models to be integrated into real-world vehicular environments.

Although existing models have advanced in predicting traffic accident severity, our work extends this line of research by jointly analyzing injuries, fatalities, and property damage within a single framework. Property damage is as important as casualties in risk assessment, particularly for vulnerable road users, so a comprehensive evaluation should consider both outcomes rather than focusing solely on injury or death. Prior studies often simplify death and property loss as binary variables, which constrains the granularity of severity analysis and limits decision support [38]. Traditional statistical models such as Logistic Regression or Ordered Probit provide interpretable coefficients and significance testing, yet their linear or parametric assumptions may miss complex interactions. Machine learning methods learn nonlinear decision boundaries and handle mixed-type variables efficiently, but they generally lack temporal awareness and rely on post hoc explanations for factor attribution. Deep learning improves representation learning and spatiotemporal modeling capacity, although it often depends on large datasets and remains difficult to interpret without dedicated tools. In contrast, multi-task learning leverages shared structure across related outcomes to capture intrinsic correlations and improve generalization, but it increases training complexity and requires principled loss weighting. Against this backdrop, we adopt a multi-task FT-Transformer that integrates deep feature learning with task sharing to model the coupled outcomes of injury, fatality, and property damage, and we pair it with SHAP-based analysis to transparently identify influential factors, thereby delivering a balanced trade-off between predictive performance, interpretability, and robustness and supporting more granular policy recommendations across severity dimensions.

C. EXPLAINABILITY OF TRAFFIC ACCIDENT SEVERITY PREDICTION

Understanding the key factors influencing traffic accident severity is fundamental for advancing predictive modeling and improving safety strategies. Traditional statistical models, such as logistic regression, offer interpretable coefficients that explicitly quantify the relationships between input variables and accident severity. In contrast, while machine learning and deep learning models have demonstrated superior predictive performance, their complex, non-linear structures often obscure the underlying decision-making process. As a result, explainability techniques are essential not only to improve model transparency but also to systematically analyze the impact of different factors on accident severity.

To address this, various explainable AI methods have been developed to enhance the interpretability of both machine learning and deep learning models. Prominent approaches include SHAP, LIME (Local Interpretable Model-Agnostic Explanations), and other model-agnostic techniques [39]. SHAP, in particular, offers a unified framework to assess feature importance by attributing the predicted outcome to the input features based on their contributions across all possible feature combinations. For instance, Aboulola used SHAP to

investigate the influence of multiple factors on traffic severity, revealing that road type and drug abuse level were particularly significant in contributing to traffic accidents [40]. Zahid et al. applied SHAP with CatBoost to interpret the factors affecting injury severity in traffic accidents, highlighting key variables such as vehicle count, type, and speed limits [41].

Recent advancements in causal inference models, particularly those integrating causal machine learning (CML), have enhanced the interpretability of traffic accident severity prediction. Causal inference methods aim to model cause–effect relationships under explicit assumptions, providing deeper insights than correlation-based approaches. For instance, Xiao and Duan combined Bayesian neural networks with causal inference to uncover causal links between accident severity and contributing factors, offering improved interpretability [42]. Similarly, Srivastava et al. used causal machine learning models to reveal key causal relationships, outperforming traditional models in both accuracy and interpretability [43]. While causal inference models provide stronger causal explanations by identifying cause-and-effect relationships, they can require more complex modeling processes and may benefit from large or well-structured datasets for accurate inference. On the other hand, SHAP offers efficient post-hoc explanations by quantifying feature contributions to model predictions, providing valuable insights into model behavior. Unlike causal inference methods, SHAP focuses on explaining correlations rather than causal mechanisms. While SHAP excels in computational efficiency and applicability, it cannot directly uncover causal relationships between factors and outcomes.

In deep learning applications, SHAP has been employed to improve model interpretability by identifying the contribution of each input feature across network layers, providing detailed insights into the factors influencing predictions. Other methods, such as DeepLIFT and Integrated Gradients, also use backpropagation to assign importance scores and highlight the impact of specific features on predictions [44]. Additionally, techniques like GCV-LIME and Sobol have been used to interpret complex models, offering alternative approaches to understanding feature importance and model behavior [45], [46].

Explainable AI methods have greatly advanced the understanding of decision-making in opaque models by revealing the underlying factors driving predictions. In this study, we apply SHAP to quantify the contributions of different factors to traffic accident severity at both the overall and sub-class levels. Rather than only assessing model transparency, we use SHAP to identify the most critical features affecting injury, fatality, and property damage severity, providing a more granular understanding of how different factors influence severity within specific sub-classes.

III. METHOD

Using accident-level counts and loss values ensures consistent measurement across years and locations and aligns with multi-task learning, which explicitly leverages the intrinsic

correlations among injury, fatality, and property-damage outcomes for joint prediction and factor attribution.

A. DATA DESCRIPTION AND PRE-PROCESSING

This study conducts an analysis based on urban road traffic accident data from Dongguan, China, spanning the years 2019 to 2023. The dataset encompasses a total of 30,442 traffic accidents that occurred during this period. The dataset includes a variety of variables related to drivers, vehicles, road conditions, and environmental factors. Some of these variables are directly extracted from traffic sensors, such as dashboard cameras, traffic monitoring cameras, and visibility sensors. These devices provide raw data concerning the causes of accidents, visibility, lighting conditions, and road status. The dataset used in this study involves sensitive personal and location-specific information and is therefore not publicly available. Anonymized excerpts may be provided upon reasonable request by contacting the corresponding author.

The dependent variables include the severity of injuries, the severity of fatalities, and the extent of property damage resulting from traffic accidents. The severity of injuries is measured by the number of individuals injured in an accident, whereas the severity of fatalities is represented by the number of deaths. Property damage is assessed by the total monetary loss, expressed in RMB. To process these three ordinal variables without losing generality, we classified them into 5, 5, and 6 categories, respectively, based on their frequency distribution, thereby converting them into ordinal variables. Table 1 presents the categories of these three traffic accident severity variables, along with their respective value ranges. The police-reported dataset provides standardized severity labels (injury/fatality/property loss) but does not include downstream medical or rehabilitation expenditure information. Consequently, cost-of-injury analysis is outside the scope of this study and will be considered in future work via linkage with hospital or insurance records.

The dataset is proprietary crash data accessed under a data-use agreement with the municipal traffic authority and, to the best of our knowledge, has not been analyzed in prior academic work for unified multi-output severity prediction. The records span 2019–2023, cover full-year seasons and all hours of day, and include driver, vehicle, environment, and roadway attributes standardized by police reporting.

Although clinical injury scales (e.g., Injury Severity Score, ISS; Abbreviated Injury Scale, AIS) are ideal for individual-level trauma assessment, such scores are not available in our large-scale accident-level database. Consistent with recent multi-task crash-severity studies, we therefore define accident-level severity as the number of injured persons, the number of fatalities, and the total monetary property loss, each discretized into ordered categories to support unified multi-task modeling and interpretation.

The independent variables in this study cover a broad spectrum of traffic accident-related characteristics, including 49 variables concerning drivers, vehicles, roads, and

TABLE 1. Categories of traffic accident severity variables.

Dependent Variable	Data Type	Number of Categories	Description
Injury severity	Ordinal	5	0, n = 0
			1, n = 1
			2, n = 2
			3, n = 3-4
			4, n >= 5
Fatal accident severity	Ordinal	5	(n = Number of injured)
			0, n = 0
			1, n = 1
			2, n = 2
			3, n = 3
Property loss severity	Ordinal	6	4, n >= 4
			(n = Number of deaths)
			0, n = 0
			1, n = 1-10
			2, n = 11-100
			3, n = 101-1000
			4, n = 1001-10000
			5, n >= 10001
			(n = Value of property loss)

environmental conditions. Table 2 provides details of all independent variables included in the dataset. Specifically, driver characteristics include variables such as gender, age, educational attainment, occupant type, driving experience, and blood alcohol concentration. Vehicle-related variables include critical information regarding the vehicle type, driving condition, turn signal and lighting status, airbag deployment, and overloading status. For road characteristics, this study examines variables such as accident location, road surface condition, traffic signal type, roadside protection facility type, pavement structure, intersection segment type, road alignment, road type, and highway administrative grade, offering a comprehensive overview of road identification, structure, and safety attributes. Environmental characteristics include variables such as weather, visibility, terrain, and lighting conditions, which are essential for analyzing the meteorological environment, visibility, geographic features, as well as the lighting conditions during the accident.

In this study, the preprocessing of traffic accident variables was meticulously conducted, considering the various characteristics of drivers, vehicles, roads, and environmental conditions, as shown in Figure 1. For driver characteristics, numerical variables such as age and driving experience are standardized to have zero mean. Categorical variables, including gender and education level, were transformed into dense embedding vectors to effectively capture the intricate relationships between different categories. Continuous variables, such as blood alcohol concentration, were normalized to a fixed range to maintain consistency across measurements.

Vehicle-related features underwent a similar preprocessing approach. Numerical variables such as vehicle age and driving status were standardized, then projected into the embedding space. Categorical variables, including vehicle

TABLE 2. Traffic accident independent variables with example values.

Independent Variable	Example values	Independent Variable	Example values
Gender	Male	Road Number	94630
Age	32	Road Name	Chashan Town Chaolang Management Area Section At the entrance of Xiayuan Alley, Chashan Town
Educational Attainment	High School	Accident Location	Motor Vehicle Lane
Occupant Type	External Worker	Road Cross-Section Position	None
Hit-and-Run Status	Yes	Central Partition	None
Driving Experience	7 years	Road Safety Attributes	Normal Road Segment
Insurance Coverage	Yes	Traffic Signal Type	None
Blood Alcohol Concentration	171 mg/100 mL	Roadside Protection	None
Seatbelt/Helmet Usage	Seatbelt used	Facility Type	None
Mode of Transport	Passenger Car	Road Physical Separation	Good
Driving Condition	Normal	Road Surface Condition	Dry
Vehicle Turn Signal Status	Not Activated	Pavement Structure	Cement
Vehicle Lighting Status	Low Beam Headlights On	Intersection Segment Type	Ordinary Road Segment
Vehicle Airbag Status	Not Deployed	Road Alignment	Straight
Overloading Status	No	Road Type	Private Community Self-Built Road
Vehicle Type	Small Car	Highway Administrative Grade	2
Motor Vehicle Condition	Normal	Road Safety Hazard Level	Low
Vehicle Legal Status	Legal	Incident Site	Dongguan City, Chashan Town
Vehicle Safety Condition	Normal	Time of Accident	2023-4-29 0:54
Hazardous Material	No	Weather	Clear
Transport Status	N/A	Visibility	More than 200 meters
Type of Hazardous Material	No	Vehicle-to-Vehicle Accident	Yes
Transported Possession of Hazardous Material	No	Terrain	Hilly
Transport Permit Number of Motor Vehicles	2	Lighting Conditions	Night with street lighting
Number of Non-Motor Vehicles	0		
Administrative Division	Dongguan City, Chashan Town		

type and legal status, were also converted into embedding vectors through the embedding layer. Binary features, such as whether the vehicle was overloaded or transporting hazardous materials, were encoded and embedded to ensure a

comprehensive representation within the model. Road characteristics, encompassing variables like surface condition, road type and safety features, were processed according to their nature—standardized or encoded—and subsequently mapped to embeddings. Environmental factors, including visibility and weather conditions, were normalized or standardized accordingly before being embedded. Time-related variables were transformed into cyclical features to reflect their periodic nature accurately and then tokenized.

To address the widely recognized differences in injury vulnerability across road user groups (e.g., pedestrians and cyclists are often more severely affected than vehicle occupants), we have categorized injured parties based on original fields such as vehicle type, mode of transport, and human role in the accident records. These fields include but are not limited to Mode of Transport (e.g., walking, e-bike, bicycle, private car), Vehicle Type (e.g., small passenger car, truck, bus), and Occupant Type (e.g., driver/passenger), which implicitly represent different road user categories (such as pedestrian, cyclist, driver, and passenger). While these groupings are not used as explicit class labels in model training, they are included as input variables (see Table 2) and serve as important contextual features to capture group-specific risk under comparable accident scenarios.

To ensure the reproducibility of the transformation pipeline from raw features to model inputs, we standardized the encoding and representation process for all feature types used in this study. All categorical variables—including administrative divisions, accident time, route identifiers, road characteristics, and attributes of vehicles and participants—were encoded using LabelEncoder fitted solely on the training split. This procedure mapped each field to a set of zero-based integer indices, with “unknown” values treated as explicit, valid categories retained in the vocabulary without any further grouping or collapsing. The resulting integer indices were subsequently passed into the Feature Tokenizer module of the FTTransformer, where they were mapped to dense vectors via embedding tables. To maintain consistent representation across heterogeneous feature types and to control model complexity, we set the embedding dimension for all categorical features to a fixed size of 32, defined by the model parameter $\text{dim}=32$, regardless of feature cardinality. Continuous and count-based variables—such as age, driving experience, weekday, and various participant or vehicle counts—were provided as floating-point inputs and projected into the same 32-dimensional latent space through a learnable linear transformation within the Feature Tokenizer. No explicit normalization or scaling was applied to continuous features, as the model learns to adapt to scale differences during training. All encoding operations and transformation parameters were exclusively fitted on the training data and held constant during validation and testing to avoid data leakage. This completes the transformation of all raw fields into unified model-consumable representations. Architectural configurations and task-specific decoding components are described in Section III-B.

B. MODEL STRUCTURE

To improve the prediction accuracy across different levels of traffic accident severity, we introduce a multi-task model, Multi-Task FT-Transformer with Uncertainty Weighting (MT-FTT-UW), which leverages the FT-Transformer architecture with uncertainty-based loss weighting, thereby enhancing the model’s overall performance, as illustrated in Figure 2. Initially, the FT-Transformer is employed to learn distinct tasks, followed by the use of uncertainty to determine the appropriate loss weights for each sub-task. Finally, interpretability is achieved through analysis with the SHAP framework.

We utilize the FT-Transformer to analyze our experimental data, focusing on three specific sub-tasks: forecasting the severity of accidents involving property damage, predicting injury-related accident severity, and estimating the severity of fatal accidents.

Figure 3 illustrates the overall architecture of the FT-Transformer model, consistently applied to the three tasks of predicting traffic accident severity. The model processes inputs in vector format, covering four categories of traffic accident features: driver, vehicle, road, and environmental attributes. Initially, these features are transformed into symbolic representations via a Feature Tokenizer, followed by the addition of a [CLS] token, which is a unique token introduced at the start of the sequence to aggregate information from all input features. This token is processed through the Transformer layers to extract a high-level representation of the entire input for subsequent learning. The [CLS] token, which stands for “classification,” is a special token added at the beginning of the input sequence. It serves as an aggregate representation of the entire input and is specifically designed to capture global information for the purpose of classification. The token’s final representation, after passing through all Transformer layers, is utilized as the input for the classification process. Essentially, the model converts all features into embeddings and processes them through multiple stacked Transformer layers [47].

1) FEATURE TOKENIZER

The Feature Tokenizer module (refer to Figure 4) transforms input features x into embeddings $E \in R^{n \times d}$ [47]. The embedding for a specific feature x_i is calculated as follows:

$$E_i = f_i(x_i) + b_i \in R^d, f_i: X_i \rightarrow R^d. \quad (1)$$

where b_i is the j -th feature bias, $f_i^{(num)}$ is implemented through the element-wise multiplication with the vector $W_j^{(num)} \in R^d$, whereas $f_i^{(cat)}$ is x_i implemented as the lookup table $W_j^{(cat)} \in R^{S_i \times d}$ for categorical features [47]. In summary:

$$\begin{aligned} E_i^{(num)} &= b_i^{(num)} + x_i^{(num)} \cdot W_i^{(num)} \in R^d, \\ E_i^{(cat)} &= b_i^{(cat)} + e_i^T \cdot W_i^{(cat)} \in R^d, \\ E &= \text{stack}[E_1^{(num)}, \dots, E_{k^{(num)}}^{(num)}, E_1^{(cat)}, \\ &\quad \dots, E_{k^{(cat)}}^{(cat)}] \in R^{n \times d}. \end{aligned} \quad (2)$$

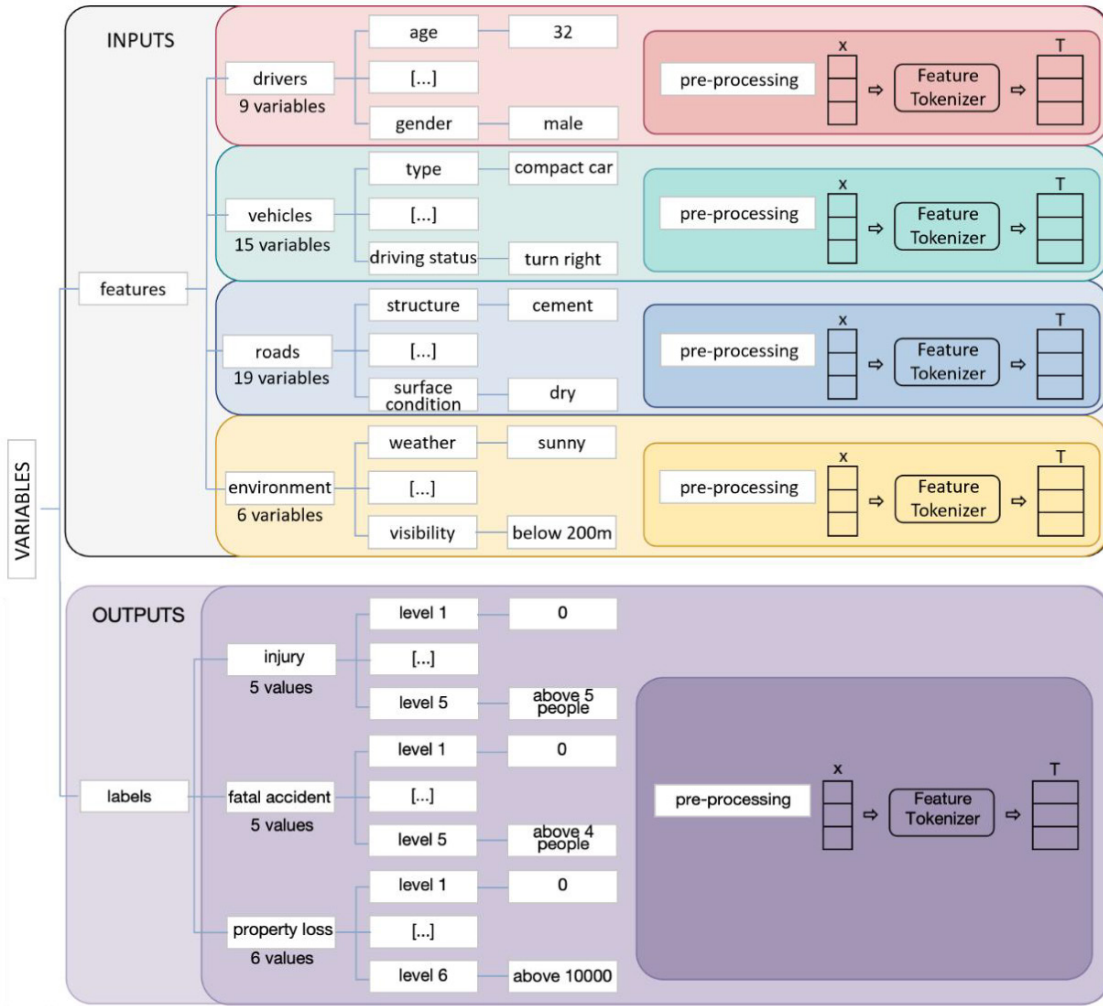


FIGURE 1. Preprocessing pipeline for traffic accident data.

where e_j^T is a one-hot vector for the corresponding categorical feature [47].

2) TRANSFORMER

At this point, the [CLS] token's embedding (also called the "prediction" or "output" token) is appended to the input sequence. This embedding is a high-dimensional vector that captures a summary of the input features, useful for classification tasks. Subsequently, E and L Transformer layers $F_1, \dots, F_L$ are applied [48]:

$$T_0 = \text{stack}[[\text{CLS}], T] \quad T_i = F_i(T_{i-1}). \quad (3)$$

We adopt the PreNorm Transformer configuration for more stable optimization (Fig. 4), following Wang et al. [47], [49]. In our implementation, we remove the initial normalization in the first Transformer layer, which yields the best validation performance. Standard details of multi-head self-attention and feed-forward blocks follow prior work [50], with additional implementation specifics provided in the supplementary materials [51].

3) PREDICTION

The final representation of the [CLS] token is used for prediction:

$$\hat{y} = \text{Linear}(\text{ReLU}(\text{LayerNorm}(T_L^{[\text{CLS}]}))). \quad (4)$$

$T_L^{[\text{CLS}]}$ denotes the [CLS] token representation from the last Transformer layer;

LayerNorm is the layer normalization operation;

ReLU is the Rectified Linear Unit activation function;

Linear is the linear projection that maps the features into the prediction space;

$\hat{y}$ is the final prediction output.

We formulate injury severity, fatal accident severity, and property-loss severity prediction as a unified multi-task learning problem to enable shared representation learning across related tasks, thereby improving generalization for each task. The overall objective is defined as a weighted sum of task-specific losses:

$$L_{\text{total}} = \sum_{i=1}^N w_i L_i \quad (5)$$

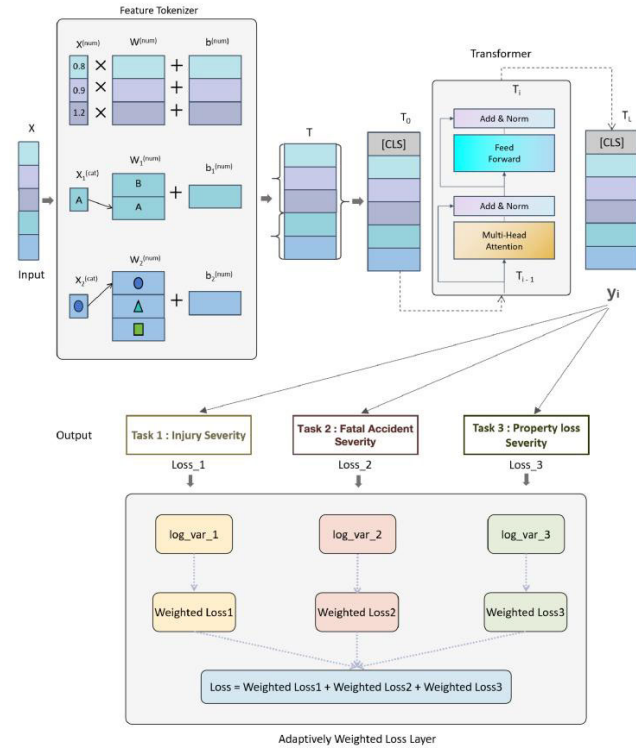


FIGURE 2. Multi-task transformer-based model with adaptive loss weighting for severity prediction.

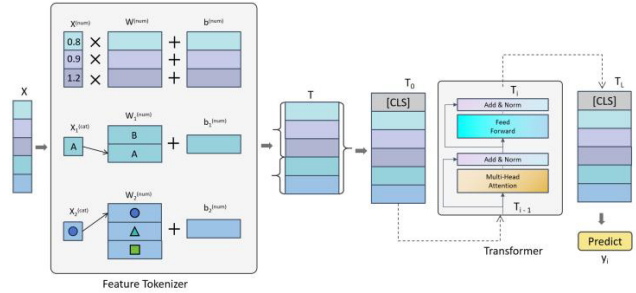


FIGURE 3. The FTT architecture.

N is the total number of tasks ($N = 3$ in this study);

L_i denotes the loss function for the i -th task;

w_i represents the weight assigned to the i -th task loss.

A fixed weighted-sum objective is sensitive to the choice of task weights, and tuning these hyperparameters is often costly and time-consuming [52]. Therefore, we prefer an approach that can automatically learn appropriate loss weights during training.

Following prior work [53], we adopt uncertainty-based loss weighting using task-dependent (homoscedastic) uncertainty. Each task is associated with a learnable uncertainty parameter that captures its relative confidence, enabling the model to adaptively balance task losses without manual weight tuning.

We derive the multi-task objective using homoscedastic uncertainty. Let $f^W(x)$ be the output of a FT-Transformer with model outputs $y_1, \dots, y_K$ (e.g., predicting property damage

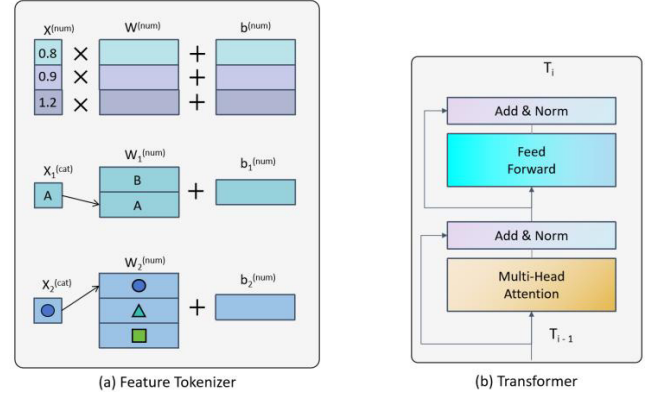


FIGURE 4. The feature tokenizer module (a) and One transformer layer.

accident severity, injury accident severity, fatal accident severity, etc.) [53]. For classification, we obtain task probabilities via a softmax function:

$$p(y|f^W(x)) = \text{Softmax}(f^W(x)) \quad (6)$$

For multiple model outputs, likelihood is commonly defined by factoring across outputs and using sufficient statistics. We define $f^W(x)$ as our sufficient statistics, and derive the following multi-task likelihood [53]:

$$p(y_1, \dots, y_K | f^W(x)) = p(y_1 | f^W(x)) \dots p(y_K | f^W(x)) \quad (7)$$

Following [53], we scale the logits with a task-dependent parameter σ :

$$p(y|f^W(x), \sigma) = \text{Softmax}\left(\frac{1}{\sigma^2} f^W(x)\right) \quad (8)$$

where $\sigma > 0$ is a scaling parameter that controls the confidence of the predicted distribution. The corresponding log-likelihood is given by [53]:

$$\log p(y=c|f^W(x), \sigma) = \frac{1}{\sigma^2} f_c^W(x) - \log \sum_{c'} \exp\left(\frac{1}{\sigma^2} f_{c'}^W(x)\right) \quad (9)$$

with $f_c^W(x)$ the c 'th element of the vector $f^W(x)$.

Assume a model with multiple outputs, y_1 and y_2 , both modeled using a softmax likelihood. As before, the joint loss, $\mathcal{K}(W, \sigma_1, \sigma_2)$, is defined as follow [53]:

$$\begin{aligned} &= -\log p(y_1, y_2 = c | f^W(x)) \\ &= -\log \text{Softmax}(y_1 = c_1; f^W(x), \sigma_1) \\ &\quad \cdot \text{Softmax}(y_2 = c_2; f^W(x), \sigma_2) \\ &= -\log p(y_1 = c_1 | f^W(x), \sigma_1) \cdot p(y_2 = c_2 | f^W(x), \sigma_2) \\ &= \frac{1}{\sigma_1^2} \mathcal{K}_1(W) + \log \frac{\sum_{c'_1} \exp\left(\frac{1}{\sigma_1^2} f_{c'_1}^W(x)\right)}{\left(\sum_{c'_1} \exp\left(\frac{1}{\sigma_1^2} f_{c'_1}^W(x)\right)\right)^{\frac{1}{\sigma_1^2}}} \end{aligned}$$

$$\begin{aligned}
& \frac{1}{\sigma_2^2} \mathcal{K}_2(W) + \log \frac{\sum_{c'_2} \exp\left(\frac{1}{\sigma_2^2} f_{c'_2}^W(x)\right)}{\left(\sum_{c'_2} \exp\left(\frac{1}{\sigma_2^2} f_{c'_2}^W(x)\right)\right)^{\frac{1}{\sigma_2^2}}} \\
& \approx \frac{1}{\sigma_1^2} \mathcal{K}_1(W) + \frac{1}{\sigma_2^2} \mathcal{K}_2(W) + \log \sigma_1 + \log \sigma_2 \quad (10)
\end{aligned}$$

where $K(W) = -\log \text{Softmax}(y_1, f^W(x))$ for the cross entropy loss of y_1 (with $f^W(x)$ not scaled), and similarly for $K_2(W)$, and optimize with respect to W as well as σ_1, σ_2 . In the last transition we introduced the explicit simplifying assumption $\frac{1}{\sigma_2} \sum_{c'_1} \exp\left(\frac{1}{\sigma_1^2} f_{c'_1}^W(x)\right) \approx \left(\sum_{c'_1} \exp\left(\frac{1}{\sigma_1^2} f_{c'_1}^W(x)\right)\right)^{\frac{1}{\sigma_2}}$ which becomes an equality when $\sigma_2 \rightarrow 1$ [53]. This approach simplifies the optimization objective and empirically improves the results.

This method accommodates arbitrary combinations of discrete or continuous loss functions and enables a principled and systematic assessment of their relative contributions. Concretely, we adopt an uncertainty-weighted objective in which task-specific loss weights are learned from predictive uncertainty and are regularized with a log-variance term to prevent weight collapse. Unlike manually chosen fixed weights, this data-driven weighting leads to stable optimization under imbalanced severity distributions and improves predictive performance for rare high-severity outcomes, which we regard as a core methodological contribution of the proposed framework.

C. MODEL EXPLAINABILITY

We analyze the contribution of input features to accident severity predictions using SHAP. SHAP estimates the marginal contribution of each feature by comparing the model's predictions across different feature combinations, providing a quantitative measure of feature importance and enhancing model transparency. SHAP offers a game theory-based framework that boosts the interpretability of machine learning models. Specifically, SHAP considers the prediction target as part of an additive feature attribution method, summing input features to show each feature's contribution to the model's predictions. For a specific sample i with its corresponding feature vector $\{x_{i1}, x_{i2}, \dots, x_{ik}, \dots, x_{in}\}$, where $k = 1, 2, \dots, n$, the model's prediction for this sample is y_i and the average prediction across all samples is denoted as $\bar{y}_i$.

$$\begin{aligned}
y_i &= \bar{y} + f(x_{i1}) + f(x_{i2}) + \dots + f(x_{ik}) + \dots + f(x_{in}), \\
k &= 1, 2, \dots, n \quad (11)
\end{aligned}$$

IV. EXPERIMENTAL RESULTS

A. EXPERIMENT SETTING

We conducted experiments using the dataset described in Section III-A, which was randomly split into three parts: 19,483 samples for training, 4,871 samples for validation, and 6,088 samples for testing. Models were trained

on the training set, fine-tuned on the validation set, and evaluated on the test set. The Adam optimizer, with a learning rate of 10-4 and a batch size of 500, was used for optimization [54]. Early stopping was implemented to halt training if validation loss did not improve after 30 epochs, helping to prevent overfitting and ensuring good generalization.

In this study, we use eight baseline models for comparison: Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), Categorical Boosting (CatBoost), a multi-layer perceptron (MLP), decision tree (DT), random forest (RF), support vector machine (SVM), and k-nearest neighbors (KNN). The MLP, with two hidden layers of 100 and 50 neurons, is trained for up to 500 iterations. The decision tree and random forest use tree-based methods, with the random forest comprising 100 trees. SVM and KNN are chosen for their distinct classification approaches, with SVM using hyperplane separation and KNN relying on distance metrics. Each model is independently trained on three tasks: predicting injury severity, fatalities, and property damage. To address class imbalance, we apply SMOTE with $k_neighbors$ set to 1 for all tasks. After splitting the dataset into training, validation, and test sets, SMOTE is applied only to the training set, while the validation and test sets remain unchanged. Data standardization is performed for MLP, SVM, and KNN; the scaler is fitted on the training set only and then applied to the validation and test sets, whereas DT, RF, XGBoost, LightGBM, and CatBoost models are trained on raw data. Models are trained on the over-sampled training data and evaluated on the original test set using accuracy, precision, recall, and F1 score. MLP, DT, RF, SVM, and KNN are implemented using scikit-learn, while XGBoost, LightGBM, and CatBoost are implemented using their respective libraries.

To quantitatively evaluate the relationships among the three target outcomes—serious injuries, slight injuries, and property loss—we conducted correlation analyses using both Pearson's correlation coefficient (r) and Spearman's rank correlation coefficient (ρ). The results indicated moderate and consistent dependencies: $r = 0.498$, $\rho = 0.502$ between serious and slight injuries, $r = 0.369$, $\rho = 0.359$ between serious injuries and property loss, and $r = 0.311$, $\rho = 0.307$ between slight injuries and property loss. These patterns reveal that while the tasks are not fully redundant, they share substantial statistical associations. This supports our multi-task learning framework, which relies on the presence of shared representations across tasks to improve generalization. Specifically, common predictive factors such as collision type, vehicle type, time of day, and rescue response time are expected to influence all three outcomes to varying degrees. The moderately correlated outcomes justify the use of shared encoder layers to extract common representations, while the task-specific decoder branches account for the nuanced distinctions (e.g., property damage is more sensitive to vehicle type, while serious injuries may depend more on crash severity). Thus, these correlation findings empirically

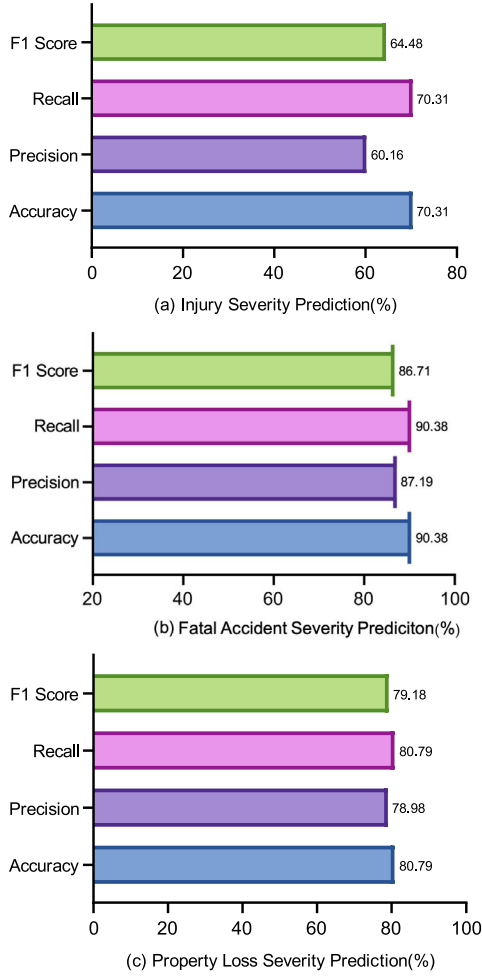


FIGURE 5. Performance metrics of the multi-task ft-transformer on traffic accident severity prediction tasks.

support our architectural choice to jointly model the three targets using a multi-task strategy.

B. EXPERIMENTS FOR PREDICTION QUALITY COMPARISON

We evaluated the performance of our traffic accident severity prediction model in terms of injuries, fatalities, and property damage using standard metrics: Accuracy (ACC), Precision (P), Recall (R), and F1 Score (F1). Below are the definitions of these metrics:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (12)$$

$$P = \frac{TP}{TP + FP} \quad (13)$$

$$R = \frac{TP}{TP + FN} \quad (14)$$

$$F1 = \frac{2PR}{P + R} \quad (15)$$

In Equations (12)–(15), the terms are defined as follows: True Positive (TP) represents the count of correctly identified positive instances. True Negative (TN) refers to the correctly

TABLE 3. Accuracy comparison of multi-task ft-transformer and baseline models on traffic accident severity prediction tasks.

Model	Injury Severity	Fatal Accident Severity	Property Loss Severity
MT-FTT-UW	0.70	0.90	0.81
XGBoost	0.66	0.85	0.75
LightGBM	0.66	0.87	0.76
CatBoost	0.65	0.84	0.71
MLP	0.58	0.83	0.68
DT	0.54	0.81	0.70
RF	0.63	0.82	0.71
SVM	0.62	0.83	0.73
KNN	0.50	0.76	0.63
MT-CNN	0.58	0.82	0.62

TABLE 4. Comparative performance of multi-task models in predicting traffic accident severity across multiple risk categories.

	Injury Severity	Fatal Accident Severity	Property Loss Severity
MT-FTT-UW	0.70	0.90	0.81
ST-FTT	0.64	0.79	0.72
MT-FTT	0.66	0.82	0.72
MT-FTT-GN	0.66	0.83	0.74

identified negative instances. False Positive (FP) counts negative instances wrongly predicted as positive. False Negative (FN) reflects positive instances incorrectly classified as negative [55].

we conducted a quantitative experiment to assess the prediction accuracy and effectiveness of the proposed framework, utilizing the traffic accident dataset from China outlined in Subsection III-A. In line with prior research [56], we evaluate the model's predictions using metrics such as ACC, P, R, and F1. ACC measures overall accuracy as the proportion of correct predictions. Precision (P) is the ratio of true positives to predicted positives, whereas Recall (R) indicates sensitivity, calculated as the ratio of true positives to actual positives. The F1 score, a harmonic mean of Precision and Recall, balances both metrics. Figure 5 presents the Multi-Task FT-Transformer model's performance on three traffic accident severity prediction tasks. For injury severity (Figure 5a), the model achieved an F1 score of 64.48%, with 70.31% for both accuracy and recall, and 60.16% for precision. In the fatal accident severity task (Figure 5b), the model performed better, with an F1 score of 86.71%, accuracy and recall at 90.38%, and precision at 87.19%. For property loss severity (Figure 5c), it attained an F1 score of 79.18%, with 80.79% for both accuracy and recall, and 78.98% for precision.

Table 3 presents a comparison of the accuracy of our MT-FTT-UW framework with five baseline models (MLP, DT, RF, SVM, and KNN) as well as three gradient boosting models (XGBoost, LightGBM, and CatBoost). As demonstrated in the results, MT-FTT-UW consistently outperforms all

TABLE 5. Case 1 - The five most influential features identified by various traffic accident severity prediction methods.

Injury Severity					
	MT-FTT-UW	DT	MLP	RF	SVM
Prediction	0(0)	2(1)	2(1)	2(2)	2(2)
1st feature	Pavement Structure (Asphalt)	Administrative Level (Municipal Level)	Type of Roadside Protection Facility (2)	Type of Roadside Protection Facility (2)	Type of Roadside Protection Facility (2)
2nd feature	Position on Road Cross-Section (Motor Vehicle Lane)	Human Category (Other)	Gender (Male)	Age (44)	Blood Alcohol Content (Unknown)
3rd feature	Time of Accident (2016-01-22 09:50:00)	Gender (Male)	Human Category (Other)	Scene (Original)	Carrying Dangerous Goods (Unknown)
4th feature	Administrative Division (Houjie Town, Dongguan City)	Age (45)	Blood Alcohol Content (0)	Gender (Male)	Property loss (500)
5th feature	Blood Alcohol Concentration (0)	Type of Roadside Protection Facility (44)	Scene (Original)	Pavement Structure (Asphalt)	Age (44)
Fatal Accident Severity					
	MT-FTT-UW	DT	MLP	RF	SVM
Prediction	0(0)	0(0)	0(0)	0(0)	0(0)
1st feature	Pavement Structure (Asphalt)	Weather (Overcast)	Type of Roadside Protection Facility (2)	Type of Roadside Protection Facility (2)	Road Surface Status (Wet)
2nd feature	Position on Road Cross-Section (Motor Vehicle Lane)	Mode of Transport (Small Passenger Vehicle)	Administrative Division (Dongguan City, Houjie Town)	Road Traffic Safety Level (Municipal Level)	Number of Pedestrians (0)
3rd feature	Time of Accident (2016-01-22 09:50:00)	Accident Occurrence Time (Unknown)	Mode of Transport (small passenger vehicle)	Central Separation Zone (4)	Blood Alcohol Content (Unknown)
4th feature	Administrative Division (Houjie Town, Dongguan City)	Road Traffic Safety Level (Unknown)	Gender (Male)	Accident Occurrence Time (Unknown)	Mode of Transport (Small Passenger Vehicle)
5th feature	Blood Alcohol Concentration (0)	Type of Roadside Protection Facility (44)	Accident Category (Injury Accident)	Mode of Transport (Small Passenger Vehicle)	Type of Roadside Protection Facility (2)
Property Loss Severity					
	MT-FTT-UW	DT	MLP	RF	SVM
Prediction	3(3)	0(1)	0(0)	0(0)	0(1)
1st feature	Pavement Structure (Asphalt)	Weather (Overcast)	Administrative Division (Dongguan City, Houjie Town)	Administrative Division (Dongguan City, Houjie Town)	Gender (Male)
2nd feature	Position on Road Cross-Section (Motor Vehicle Lane)	Human Category (Other)	Gender (Male)	Gender (Male)	Road Physical Isolation (No partition)
3rd feature	Time of Accident (2016-01-22 09:50:00)	Administrative Division (Dongguan City, Houjie Town)	Number of Casualties (1)	Number of Casualties (1)	Road Type (Straight)

TABLE 5. (Continued.) Case 1 - The five most influential features identified by various traffic accident severity prediction methods.

4th feature	Day of the Week (5)	Mode of Transport (Small Passenger Vehicle)	Scene (Original)	Scene (Original)	Age (44)
5th feature	Administrative Division (Houjie Town, Dongguan City)	Gender (Male)	Accident Occurrence Time (2016-01-22 09:50:00)	Human Category (None)	Lighting Conditions (Daylight)

competing methods across the three traffic accident severity prediction tasks. For injury severity, MT-FTT-UW achieves an accuracy of 70%, surpassing SVM by 8% and outperforming MLP, DT, RF, and KNN by 12%, 16%, 7%, and 20%, respectively. It also exceeds XGBoost, LightGBM, and CatBoost by 4%, 4%, and 5%, respectively. In the fatal accident task, MT-FTT-UW reaches 90% accuracy, outperforming SVM by 7% and MLP, DT, RF, and KNN by 7%, 9%, 8%, and 14%, respectively, while further surpassing XGBoost, LightGBM, and CatBoost by 5%, 3%, and 6%. For property loss severity, MT-FTT-UW achieves 81% accuracy, outperforming SVM by 8% and MLP, DT, RF, and KNN by 13%, 11%, 10%, and 18%, respectively, and also exceeding XGBoost, LightGBM, and CatBoost by 6%, 5%, and 10%. These results demonstrate that MT-FTT-UW maintains consistent superiority even when compared against strong gradient boosting baselines.

When compared with the MT-CNN model proposed by Yang et al., MT-FTT-UW also shows clear advantages across all three prediction tasks. Specifically, MT-FTT-UW achieves accuracy scores of 0.70, 0.90, and 0.81 for injury severity, fatal accident severity, and property loss severity, respectively, whereas MT-CNN attains 0.58, 0.82, and 0.62 on the same tasks. This highlights MT-FTT-UW's advanced feature extraction capabilities, particularly suited for high-dimensional, nonlinear, time-series data, reinforcing its strength within a multi-task learning framework.

To evaluate the effectiveness of the proposed MT-FTT-UW, three comparisons are conducted in this section. First, MT-FTT-UW is compared with a Single-Task FT-Transformer (ST-FTT) to assess the benefits of multi-task learning. Second, MT-FTT-UW is compared with the standard Multi-Task FT-Transformer (MT-FTT) without uncertainty weighing to determine the impact of incorporating uncertainty in loss weighting. Finally, MT-FTT-UW is compared against the Multi-Task FT-Transformer using GradNorm (MT-FTT-GN) to assess the effectiveness of different optimization strategies [57]. These comparisons highlight the contributions of multi-task learning, the FT-Transformer architecture, and the specific optimization techniques to the overall performance of the MT-FTT-UW model. The results are shown in Table 4.

When comparing the MT-FTT-UW model to the ST-FTT, MT-FTT-UW consistently outperforms ST-FTT across all tasks: Injury Severity, Fatal Accident Severity, and Property Loss Severity. In Injury Severity, MT-FTT-UW scores 0.70, compared to 0.64 for ST-FTT. In Fatal Accident Severity, it scores 0.90, surpassing ST-FTT's 0.79. Finally, in Property

Loss Severity, it achieves 0.81, a notable improvement over ST-FTT's 0.72. This consistent improvement highlights the benefits of multi-task learning, enabling the model to capture broader features by sharing information across tasks. This approach enhances performance across multiple tasks, suggesting that multi-task learning can significantly boost generalization in complex, interrelated tasks.

Compared to MT-FTT, MT-FTT-UW shows superior performance. In Property Loss Severity, MT-FTT-UW scores 0.81, surpassing MT-FTT's 0.72, showing the benefit of uncertainty weighting. It also outperforms MT-FTT in Injury Severity (0.70 vs. 0.66) and Fatal Accident Severity (0.90 vs. 0.82). Uncertainty weighting helps the model focus more effectively, reducing task interference and improving overall performance.

When compared with MT-FTT-GN, MT-FTT-UW remains highly competitive, especially in Property Loss Severity. This suggests that whereas GradNorm improves performance in specific tasks, MT-FTT-UW's uncertainty-based approach offers a more balanced improvement across tasks. This balance is crucial in multi-task learning, as it ensures the model maintains equilibrium without disproportionately optimizing certain tasks at the expense of others.

In conclusion, the MT-FTT-UW model shows how uncertainty-based loss weighting within the FT-Transformer architecture enhances multi-task learning, particularly in predicting traffic accident risk levels. By enabling information sharing across interrelated tasks like injury, fatality, and property damage severity, the model captures the connections between different severity types, which is crucial in traffic accident contexts. This shared learning improves prediction accuracy across all severity levels, enabling the model to outperform baseline approaches and minimize task interference, especially in injury severity prediction. Ultimately, this approach offers a balanced risk profile, optimizing multi-task performance and addressing common trade-offs in multi-task learning.

C. MULTI-LEVEL EXPLAINABILITY ANALYSIS: CASE STUDIES AND COHORT-LEVEL SHAP

Drawing from previous research, we randomly selected two case studies to showcase the interpretability of our proposed model [45].

Traditional interpretable models for discrete outcomes have been widely used in traffic accident severity prediction. This study compares the interpretability of the MT-FTT-UW model using SHAP analysis with conventional methods.

TABLE 6. Case 2 - The five most influential features identified by various traffic accident severity prediction methods.

Injury Severity					
	SHAP	DT	MLP	RF	SVM
Prediction	1(1)	1(0)	1(1)	1(1)	1(0)
1st feature	Pavement Structure (Asphalt)	Administrative Division (Dongguan City, Guancheng District)	Type of Roadside Protection Facility (1)	Type of Roadside Protection Facility (1)	Roadside Protection Facility Type (1)
2nd feature	Administrative Division (Dongguan City, Guancheng District)	Human Category (Self-employed)	Gender (Female)	Age (34)	Blood Alcohol Concentration (1)
3rd feature	Blood Alcohol Concentration (1)	Gender (Female)	Human Category (2)	Gender (Female)	Carrying Dangerous Goods (Unknown)
4th feature	Position on Road Cross-Section (Motor Vehicle Lane)	Age (34)	Blood Alcohol Content (1)	Blood Alcohol Content (1)	Road Surface Status (Dry)
5th feature	Carrying Dangerous Goods (2)	Type of Roadside Protection Facility (1)	Scene (Vehicular hit-and-run)	Pavement Structure (Asphalt)	Age (34)
Fatal Accident Severity					
	MT-FTT-UW	DT	MLP	RF	SVM
Prediction	0(0)	1(1)	0(0)	0(0)	0(0)
1st feature	Pavement Structure (Asphalt)	Weather (Sunny)	Type of Roadside Protection Facility (1)	Type of Roadside Protection Facility (1)	Road Surface Status (Dry)
2nd feature	Blood Alcohol Concentration (1)	Mode of Transport (Small Passenger Vehicle)	Administrative Division (Dongguan City, Guancheng District)	Road Traffic Safety Level (Municipal level)	Number of Pedestrians (2)
3rd feature	Position on Road Cross-Section (Motor Vehicle Lane)	Accident Occurrence Time (Unknown)	Mode of Transport (Small Passenger Vehicle)	Central Separation Zone (4)	Blood Alcohol Concentration (Unknown)
4th feature	Vehicle-to-Vehicle Accident (Side collision)	Road Traffic Safety Level (Municipal level)	Gender (Female)	Accident Occurrence Time (Unknown)	Mode of Transport (Small Passenger Vehicle)
5th feature	Administrative Division (Dongguan City, Guancheng District)	Type of Roadside Protection Facility (1)	Blood Alcohol Concentration (1)	Mode of Transport (Small Passenger Vehicle)	Roadside Protection Facility Type (1)
Property Loss Severity					
	MT-FTT-UW	DT	MLP	RF	SVM
Prediction	4(4)	1(0)	0(1)	0(0)	0(1)
1st feature	Pavement Structure (Asphalt)	Weather (Sunny)	Administrative Division (Dongguan City, Guancheng District)	Administrative Division (Dongguan City, Guancheng District)	Gender (Male)
2nd feature	Road Name (Qifeng Road, Urban Area, Dongguan City)	Human Category (Self-employed)	Gender (Females)	Gender (Females)	Road Physical Isolation (Central partition)

TABLE 6. (Continued.) Case 2 - The five most influential features identified by various traffic accident severity prediction methods.

3rd feature	Blood Alcohol Concentration (1)	Administrative Division (Dongguan City, Guancheng District)	Road Surface Status (Dry)	Number of Casualties (2)	Road Type (Straight)
4th feature	Position on Road Cross-Section (Motor Vehicle Lane)	Mode of Transport (Small Passenger Vehicle)	Accident Occurrence Time (2023-01-25 02:43:00)	Scene (Vehicular hit-and-run)	Age (34)
5th feature	Road Traffic Safety Level (Municipal level)	Gender (Female)	Blood Alcohol Concentration (1)	Human Category (Self-employed)	Lighting Conditions (Illuminated by streetlights at night)

TABLE 7. The five most influential features identified by various traffic accident severity prediction methods.

Injury Severity			
	SHAP	LIME	Sobol
1st feature	Pavement Structure	Position on Road Cross-Section	Gender
2nd feature	Position on Road Cross-Section	Driving Status	Vehicle Turn Signal Status
3rd feature	Time of Accident	Pavement Structure	Time of Accident
4th feature	Administrative Division	Vehicle Legal Status	Driving Status
5th feature	Blood Alcohol Concentration	Seatbelt/Helmet Usage	Pavement Structure
Fatal Accident Severity			
	SHAP	LIME	Sobol
1st feature	Pavement Structure	Road Alignment	Age
2nd feature	Position on Road Cross-Section	Vehicle Safety Condition	Road Name
3rd feature	Time of Accident	Vehicle Legal Status	Position on Road Cross-Section
4th feature	Administrative Division	Scene	Road Surface Condition
5th feature	Blood Alcohol Concentration	Vehicle Lighting Status	Gender
Property Loss Severity			
	SHAP	LIME	Sobol
1st feature	Pavement Structure	Road Alignment	Position on Road Cross-Section
2nd feature	Position on Road Cross-Section	Vehicle Legal Status	Road Physical Isolation
3rd feature	Time of Accident	Lighting Conditions	Intersection Segment Type
4th feature	Day of the Week	Administrative Division	Lighting Conditions
5th feature	Administrative Division	Traffic Signal Method (Control)	Seatbelt/Helmet Usage

Tables 5 and 6 show the five most influential features affecting severity levels in two accident cases, identified by the MT-FTT-UW model and four baseline models. The predicted

severity levels are classified according to the criteria in Table 1. In this analysis, key factors influencing the MT-FTT-UW model’s predictions are derived using SHAP.

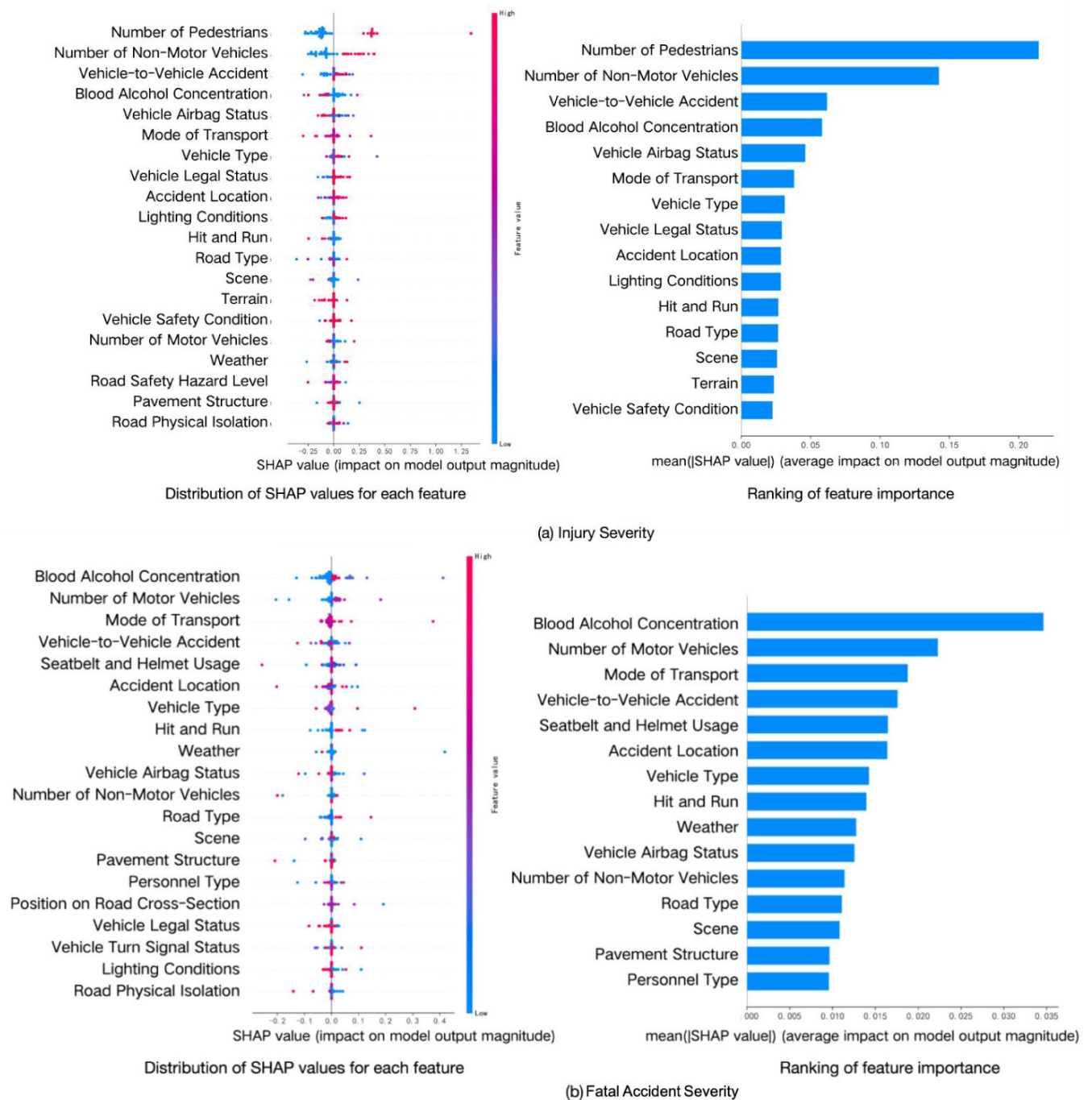


FIGURE 6. SHAP summary plot for (a) Injury severity, (b) Fatal accident severity, and (c) property loss severity.

Feature importance in Random Forest is assessed by the reduction of impurity, typically Gini impurity, during splits [58]. For MLP, it is calculated by summing the absolute weight values between the input and hidden layers [59]. In SVM, feature importance is inferred from the average absolute values of the weight coefficients in the decision function [60]. The calculation of feature importance in Decision Trees is similar to that in Random Forest.

Tables 5 and 6 show the five key features contributing to the traffic accident severity outcomes in two cases, as identified by the MT-FTT-UW model and four baseline models. In both cases, our proposed MT-FTT-UW model accurately predicts all three types of traffic accident severity, whereas the baseline methods (T, MLP, RF, and SVM) fail to predict one or two severity types. This demonstrates the robustness of the MT-FTT-UW model in handling complex, multi-dimensional data that involve various severities. Furthermore, compared

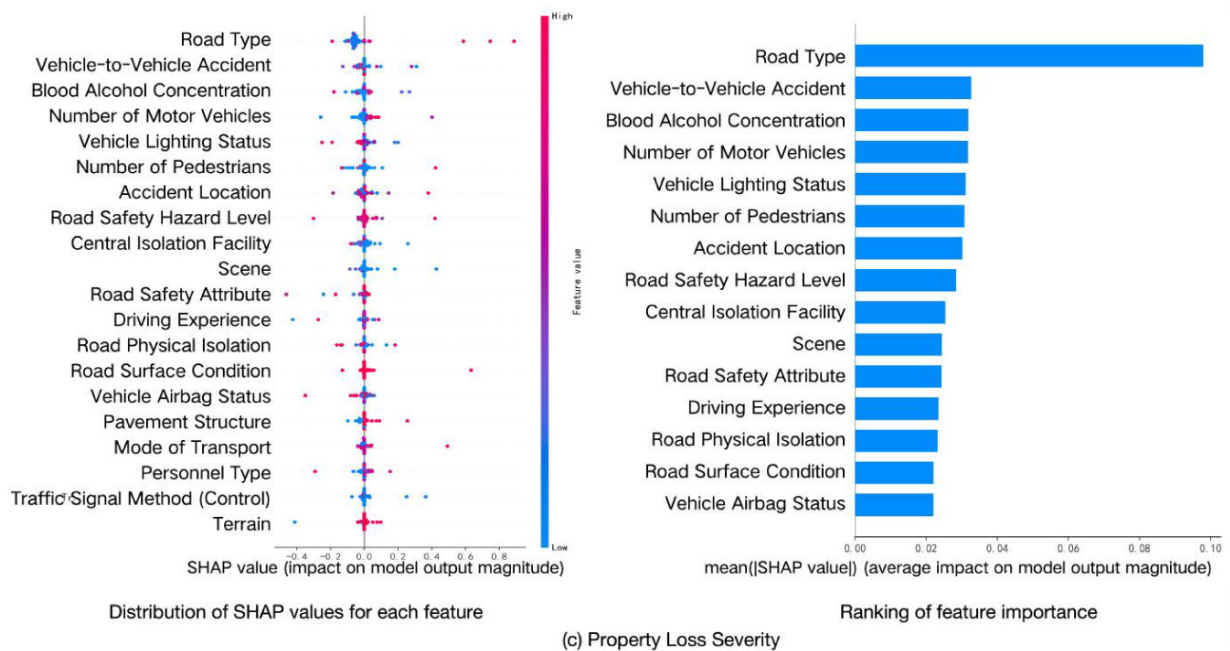


FIGURE 6. (Continued.) SHAP summary plot for (a) Injury severity, (b) Fatal accident severity, and (c) property loss severity.

to the explanations provided by the baseline models, the MT-FTT-UW model's top five features have fewer missing values. Since missing values cannot contribute to predicting traffic accident severity, this makes the MT-FTT-UW model's explanations more reasonable than those of the baseline methods. Both RF and SVM tend to introduce irrelevant or incomplete feature explanations, often leading to missing values that significantly reduce the reliability of their predictions. In contrast, the MT-FTT-UW model provides more precise explanations tailored to specific accident scenarios, better reflecting real factors like road conditions and accident timing, thus improving prediction accuracy.

The quality of explanations generated by our SHAP method is evaluated and compared with other techniques for interpreting neural networks. Specifically, LIME, used for explaining traffic accident severity predictions [61], and the Sobol method, commonly applied in sensitivity analysis, are chosen as baseline models [62]. LIME and Sobol are implemented using the Python packages lime and SALib, respectively. Table 7 presents the top five features identified by the MT-FTT-UW model using these three explanation techniques. The results indicate that both LIME and the Sobol method have a higher incidence of features with missing values compared to SHAP. Missing values could not predict traffic accident severities as they represent no information [7]. These missing values can limit the effectiveness of LIME and Sobol in providing comprehensive information about feature importance, thereby affecting their ability to accurately predict traffic accident severities. As a result, the explanations provided by LIME and Sobol provide less reliability and precision compared to SHAP. This analysis suggests that

SHAP provides more robust and informative explanations for the MT-FTT-UW model in comparison to both LIME and Sobol.

The MT-FTT-UW model was applied to predict injury, death, and property damage severity, leveraging both global and sub-class SHAP analyses to interpret model predictions. Global SHAP analysis identifies the most significant factors influencing accident severity across the entire dataset, while sub-class SHAP analysis provides finer-grained insights into the specific impact of each feature on different severity categories, highlighting variations that may not be apparent in a global analysis. Table 8 presents the top five important features for each class, and Figure 6 visualizes their SHAP value distributions. This combined approach helps clarify how different risk factors contribute to accident severity under various conditions.

Across all tasks, the most prominent common features are time of accident, vehicle airbag deployment status, and road structure and condition, especially in predicting injury severity and property damage. The time of the accident influences traffic conditions, congestion, and lighting, which directly affect accident outcomes [13]. This feature is significant across all severity classes due to its effect on driver visibility and reaction times, as well as on the density of traffic at different times. Notably, airbag deployment reflects the intensity of the impact rather than the presence or effectiveness of airbag systems. Since airbags are standard equipment in most modern vehicles, their deployment is typically triggered when the crash force surpasses a certain threshold, especially in frontal collisions. Therefore, its prominence in predicting injury and fatal accident severity likely captures the underlying crash

TABLE 8. Top five features for injury, death, and property severity prediction across different classes.

Injury Severity					
	1st feature	2nd feature	3rd feature	4th feature	5th feature
Class 0	Time of Accident	Blood Alcohol Concentration	Non-Motor Vehicles	Road Number	Administrative Division
Class 1	Blood Alcohol Concentration	Time of Accident	Motor Vehicles	Administrative Division	Road Name
Class 2	Lighting Conditions	Cultural Level	Vehicle Airbag Status	Time of Accident	Vehicle-to-Vehicle Accident
Class 3	Lighting Conditions	Time of Accident	Road Name	Vehicle-to-Vehicle Accident	Road Number
Class 4	Lighting Conditions	Vehicle Airbag Status	Time of Accident	Vehicle-to-Vehicle Accident	Road Number
Fatal Accident Severity					
	1st feature	2nd feature	3rd feature	4th feature	5th feature
Class 0	Vehicle Airbag Status	Time of Accident	Lighting Conditions	Road Number	Road Condition
Class 1	Vehicle Airbag Status	Lighting Conditions	Time of Accident	Road Number	Cultural Level
Class 2	Vehicle Airbag Status	Road Condition	Time of Accident	Accident Location	Administrative Division
Class 3	Vehicle Airbag Status	Road Condition	Time of Accident	Accident Location	Accident Location
Class 4	Vehicle Airbag Status	Road Condition	Time of Accident	Personnel Type	Accident Location
Property Severity					
	1st feature	2nd feature	3rd feature	4th feature	5th feature
Class 0	Road Condition	Time of Accident	Accident Location	Road Number	Scene
Class 1	Vehicle Airbag Status	Vehicle-to-Vehicle Accident	Accident Location	Road Condition	Accident Location
Class 2	Vehicle Airbag Status	Vehicle-to-Vehicle Accident	Time of Accident	Lighting Conditions	Road Number
Class 3	Blood Alcohol Concentration	Road Condition	Administrative Division	Time of Accident	Personnel Type
Class 4	Blood Alcohol Concentration	Administrative Division	Road Number	Time of Accident	Vehicle Type
Class 5	Vehicle Airbag Status	Time of Accident	Vehicle-to-Vehicle Accident	Time of Accident	Road Number

energy rather than a protective mechanism. It is important to clarify that this variable serves as a contextual signal of severe impact events, not a causal factor in injury reduction. Conversely, airbag deployment has minimal direct relevance to property damage severity, as its activation is designed to protect occupants rather than prevent vehicle damage. Road structure and conditions become more important as severity increases. Complex road conditions, such as wet surfaces or curves, contribute to higher severity by compromising vehicle stability and control [13]. These conditions are key predictors of accidents with higher property damage and fatality rates, as they make vehicle handling and control more challenging.

Feature dependence varies across tasks. “Blood alcohol concentration (BAC)” is a key predictor for injury severity, reflecting how driver impairment worsens accident outcomes

[65]. Whereas BAC influences injury severity, it is less relevant for predicting death or property damage, where factors like collision location, vehicle speed, and structural integrity are more important. For Fatal Accident Severity, collision location and administrative division emerge as significant predictors, highlighting the role of infrastructure and traffic patterns in accident risks [66]. Specific areas may have a history of high-severity accidents due to environmental or design factors.

For property damage severity, vehicle type and accident location are key factors. Certain vehicle types incur higher repair costs, and accidents at busy intersections or narrow roads tend to result in more significant damage due to limited maneuverability [67]. Administrative divisions, reflecting regional traits, are also important for predicting death and

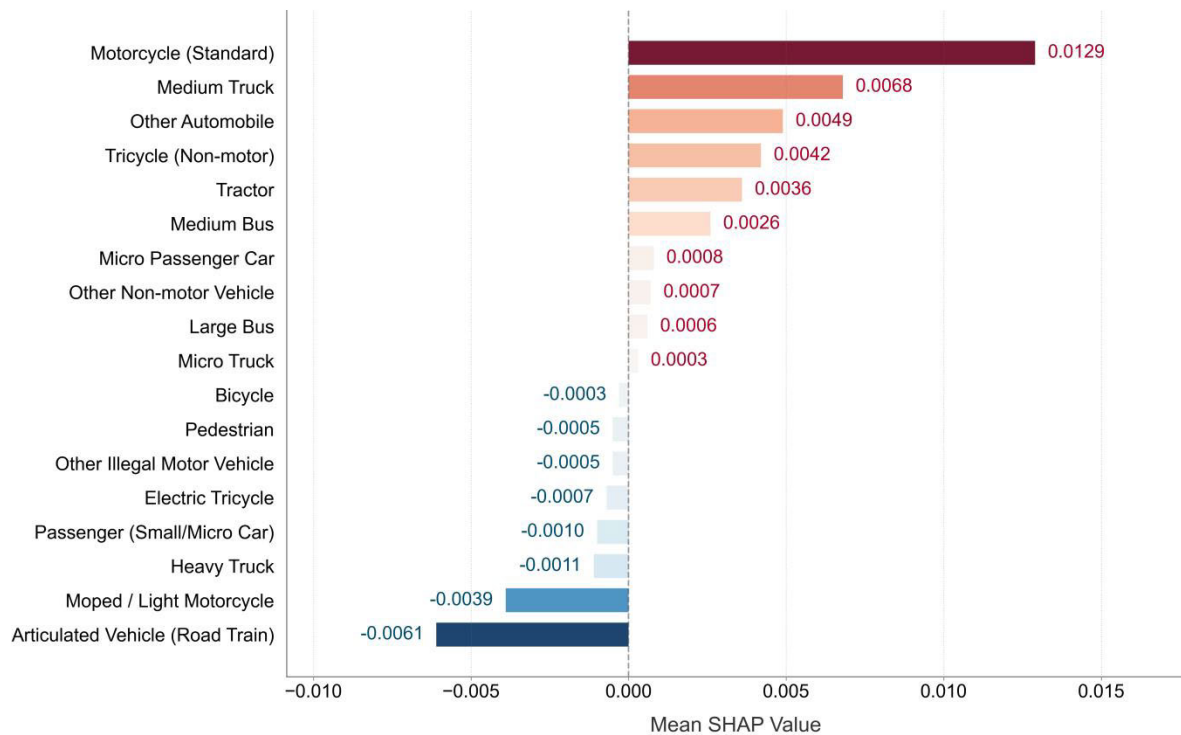


FIGURE 7. Cohort-level SHAP analysis of participant category: mean SHAP contributions to the predicted injury risk.

property damage severity, suggesting that tailored regional safety policies are needed based on local road conditions and driving behavior.

Methodologically, we extend SHAP to a sub-class resolution within a multi-output model, enabling class-specific attribution that translates into actionable, targeted interventions (e.g., lighting/visibility for fatality classes vs. vehicle mix and collision configuration for high property-loss classes). Empirically, our sub-class SHAP analysis reveals that the three severity outcomes are governed by distinct but partially overlapping mechanisms: injury severity is primarily driven by driver impairment and visibility, fatal accident severity is strongly associated with impact energy and road geometry, and property-loss severity is largely influenced by vehicle type and collision structure. This heterogeneity has not been systematically demonstrated in prior traffic safety studies and thus constitutes new empirical evidence on the differentiated formation of accident severity.

To quantify the independent effect of participant category on injury severity under otherwise identical conditions, a cohort-level SHAP analysis was conducted. As shown in Fig. 7, negative SHAP values indicate a reduction in the predicted injury risk relative to the model baseline, whereas positive values indicate an increase. Several participant categories exhibit negative SHAP contributions, including vehicle trains (-0.0061), light motorcycles (-0.0039), heavy trucks (-0.0011), small passenger vehicles (-0.0010), and three-wheeled electric vehicles (-0.0007). These negative

contributions may be attributed to structural characteristics such as enclosed cabins, lower operating speeds, or increased vehicle mass, which can mitigate occupant injury risk once other crash-related factors are controlled. Pedestrians (-0.0005) and bicycles (-0.0003) also show small negative SHAP values. This should not be interpreted as pedestrians being intrinsically safer. Instead, it reflects a conditional effect after controlling for other crash-related factors. Meanwhile, multiple categories present SHAP values equal or very close to zero (e.g., small passenger cars, shared bicycles, rental bicycles, low-speed vehicles, and special-purpose vehicles). These near-zero contributions suggest that, after accounting for dominant variables such as impact severity, lighting conditions, and alcohol involvement, participant category alone provides limited additional explanatory power for injury risk prediction.

In contrast, a subset of participant categories exhibits positive SHAP contributions, indicating an increased predicted injury risk. Riding a standard motorcycle shows the largest average SHAP value ($+0.0129$), followed by medium-sized trucks ($+0.0068$), other automobiles ($+0.0049$), three-wheeled vehicles ($+0.0042$), tractors ($+0.0036$), medium-sized passenger buses ($+0.0026$), and micro passenger vehicles ($+0.0008$). These positive contributions may reflect a combination of factors, such as reduced physical protection, higher exposure during collisions, or vehicle configurations that are more sensitive to crash dynamics, once other covariates are held constant. Other categories, including micro

trucks (+0.0003), large passenger buses (+0.0006), and other non-motorized vehicles (+0.0007), display positive but relatively small SHAP values, suggesting only a modest increase in injury risk. Overall, while participant categories differ in both direction and magnitude of SHAP contributions, the absolute effect sizes remain moderate, indicating that participant category functions as a secondary risk modifier rather than a dominant determinant of injury severity.

V. DISCUSSION AND CONCLUSION

In this study, we introduced a multi-task FT-Transformer framework for predicting traffic accident severity across three categories: injury, fatality, and property loss severity. By employing multi-task learning within the FT-Transformer architecture, the model is able to share informative representations across related tasks, which is essential for capturing the intrinsic relationships among different severity outcomes. In particular, this study formulates accident severity prediction as a unified multi-output learning problem within an application-driven setting, allowing the model to explicitly encode and utilize the statistical dependencies among injuries, fatalities, and property-loss outcomes, rather than treating them as isolated targets. Additionally, we applied an uncertainty-weighted loss function to dynamically balance task-specific contributions during training, improving the model's robustness in the presence of class imbalance and rare high-severity cases. Compared with conventional multi-task models using fixed or manually tuned weights, our approach demonstrates consistently higher predictive accuracy and more balanced performance across severity levels, especially when handling complex, heterogeneous traffic accident data.

Our model also emphasizes interpretability, utilizing SHAP (Shapley Additive Explanations) to identify and analyze the key factors influencing accident severity across each sub-class level within the multi-class classification. This approach allows us to distinguish the unique contributing factors that impact varying levels of injury, fatality, and property damage. By applying SHAP analysis to each severity sub-class, the model provides detailed insights into how specific variables (such as driver characteristics, environmental conditions, vehicle type, and road structure) differentially affect the probability and severity of accidents. This sub-class-level interpretability enables practitioners and policymakers to understand the distinct underlying causes of different severity outcomes, supporting tailored safety interventions that address the specific risk factors associated with each severity level. This detailed interpretability not only enhances transparency in decision-making but also strengthens the model's reliability and practical applicability for traffic safety management.

In terms of practical implications, the insights from our model can guide the development of traffic safety policies, particularly for VRUs and smart city initiatives. By accurately predicting accident severity and identifying key factors, urban planning and transportation departments can proactively

mitigate risks in high-risk areas, reducing casualties and property loss. The model's interpretability allows for targeted interventions, such as improving road conditions or enhancing infrastructure where needed. Additionally, by integrating these predictions into intelligent transportation systems, cities can improve real-time traffic management and enhance public safety in a more adaptive and sustainable way [68].

In practice, MT-FTT-UW supports two uses: offline planning and online operations. Offline, agencies create joint risk maps and hotspot lists by month or quarter. Online, the model turns multi-class outputs into tiered alerts and dispatch guidance. Test accuracies are 70.31% for injury, 90.38% for death, and 80.79% for property loss. SHAP highlights time of accident, lighting, airbag deployment, and road structure or condition, which map to actions such as lighting upgrades at night, wet-weather speed control and surface or drainage fixes. The uncertainty-weighted multitask design keeps tasks balanced.

The multi-task FT-Transformer model presented for traffic accident severity prediction has certain limitations. First, its reliance on structured tabular data may limit applicability in scenarios where more complex data types, such as images or real-time sensor data, are necessary for accurate severity assessment. Incorporating unstructured data could potentially improve model robustness and generalizability. Additionally, whereas the uncertainty-weighted loss function helps balance contributions across tasks, it may not fully capture the non-linear dependencies among severity types. Additionally, although severity levels are ordinal in nature, the model adopts a categorical softmax cross-entropy formulation without explicitly modeling the ordering. Training and inference efficiency comparisons between the proposed FT-Transformer framework and tree-based baselines (e.g., XGBoost) are beyond the scope of this study.

Moreover, this study defines injury severity at the accident level due to data constraints. Clinical scales such as the ISS and AIS were unavailable in our dataset. We therefore used the number of injured persons as a proxy to ensure consistent measurement across years and locations and to align with joint modeling of fatalities and property loss. Future work will link accident records with trauma registries to validate and, where feasible, replace this proxy with clinical severity scores.

Finally, the analysis is based on police-reported crash records from a single Chinese city (Dongguan). As a result, the estimated relationships and the learned multi-task representations may exhibit regional specificity, reflecting local roadway design, traffic composition, enforcement practices, and driver behavior. The reported model performance should therefore be interpreted as in-sample performance on Dongguan data rather than a guaranteed level in other regions. Caution is warranted when extrapolating these findings to other cities or countries, and external validation with multi-city or cross-regional datasets is needed to further assess the robustness and transferability of the proposed framework.

Looking ahead, there are several promising directions for future research. First, incorporating richer data sources, such as images of road conditions and vehicle damage, could further enhance the model's accuracy and robustness. Second, improving the explainability of deep neural networks by developing methods to extract decision rules could provide more actionable insights. Finally, further advancements in multi-task learning techniques may enable even more effective sharing of information across different accident severity prediction tasks, providing more comprehensive and accurate safety solutions.

REFERENCES

- [1] M. Manzoor, M. Umer, S. Sadiq, A. Ishaq, S. Ullah, H. A. Madni, and C. Bisogni, "RFCNN: Traffic accident severity prediction based on decision level fusion of machine and deep learning model," *IEEE Access*, vol. 9, pp. 128359–128371, 2021.
- [2] *China Statistical Yearbook*, State Stat. Bureau, Zhongguo Tongji Chubanshe, Beijing, China, 2020.
- [3] M. M. Hamed and B. M. Al-Eideh, "An exploratory analysis of traffic accidents and vehicle ownership decisions using a random parameters logit model with heterogeneity in means," *Analytic methods accident Res.*, vol. 25, Jun. 2020, Art. no. 100116.
- [4] S. M. E. Shaik, I. M. M. Islam, and H. Q. S. Hossain, "A review on neural network techniques for the prediction of road traffic accident severity," *Asian Transp. Stud.*, vol. 7, Dec. 2021, Art. no. 100040.
- [5] P. Infante, G. Jacinto, A. Afonso, L. Rego, V. Nogueira, P. Quaresma, J. Saias, D. Santos, P. Nogueira, M. Silva, R. Pisco Costa, P. Gois, and P. R. Manuel, "Comparison of statistical and machine-learning models on road traffic accident severity classification," *Computers*, vol. 11, no. 5, p. 80, May 2022.
- [6] K. Bucsuházy, E. Matuchová, and R. Závada, "Human factors contributing to the road traffic accident occurrence," *Transp. Res. Proc.*, vol. 45, pp. 555–561, Jan. 2020.
- [7] Z. Yang, W. Zhang, and J. Feng, "Predicting multiple types of traffic accident severity with explanations: A multi-task deep learning framework," *Saf. Sci.*, vol. 146, Jun. 2022, Art. no. 105522.
- [8] K. Santos, J. P. Dias, and C. Amado, "A literature review of machine learning algorithms for crash injury severity prediction," *J. Saf. Res.*, vol. 80, pp. 254–269, Feb. 2022.
- [9] R. Yu, Y. Wang, and Z. Zou, "Convolutional neural networks with refined loss functions for the real-time crash risk analysis," *Transp. Res. C, Emerg. Technol.*, vol. 119, May 2020, Art. no. 102740.
- [10] S. Vandenheide, S. Georgoulis, W. Van Gansbeke, M. Proesmans, D. Dai, and L. Van Gool, "Multi-task learning for dense prediction tasks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 7, pp. 3614–3633, Jul. 2022.
- [11] W. Sun et al., "Cost-effective system for rural roadway traffic, surface conditions and weather conditions monitoring," Center Saf. Equity Transp. (CSET), Univ. Alaska Fairbanks, Fairbanks, AK, USA, Tech. Rep. INE/CSET 24.14, Jun. 2024.
- [12] K. Sherifdeen and S. Daniel, "Explainable artificial intelligence for interpreting and understanding diabetes prediction models," *EasyChair Preprint* 13785, Jan. 2024.
- [13] L. Ebohi, C. Forciniti, and G. Mazzulla, "Factors influencing accident severity: An analysis by road accident type," *Transp. Res. Proc.*, vol. 47, pp. 449–456, Jan. 2020.
- [14] M. Liang, Y. Zhang, G. Qu, Z. Yao, M. Min, T. Shi, L. Duan, and Y. Sun, "Epidemiology of fatal crashes in an underdeveloped city for the decade 2008–2017," *Int. J. Injury Control Saf. Promotion*, vol. 27, no. 2, pp. 253–260, Apr. 2020.
- [15] M. Bakhtiyari, M. R. Mehmandar, M. Khezeli, A. Latifi, T. A. Jouybari, and M. A. Mansournia, "Estimating the avoidable burden and population attributable fraction of human risk factors of road traffic injuries in iran: Application of penalization, bias reduction and sparse data analysis," *Int. J. Injury Control Saf. Promotion*, vol. 26, no. 4, pp. 405–411, Oct. 2019.
- [16] F. Hafeez, U. U. Sheikh, S. Al-Shammari, M. Hamid, A. B. K. Khakwani, and Z. A. Arfeen, "Comparative analysis of influencing factors on pedestrian road accidents," *Bull. Electr. Eng. Informat.*, vol. 12, no. 1, pp. 257–267, Feb. 2023.
- [17] M. Behzadi Goodari, H. Sharifi, P. Dehesh, M. A. Mosleh-Shirazi, and T. Dehesh, "Factors affecting the number of road traffic accidents in Kerman province, Southeastern Iran (2015–2021)," *Sci. Rep.*, vol. 13, no. 1, p. 6662, Apr. 2023.
- [18] M. Rahmaninezhad Asil, H. Toroghi, and I. Bargegol, "Analysis of factors associated with traffic injury severity on urban roads in different lighting conditions," *Comput. Res. Prog. Appl. Sci. Eng.*, vol. 8, no. S1, pp. 1–6, 2022.
- [19] N. Wijekoon, "Assessing visual impacts of roadscape installations on urban safety and comfort in Galle Road, Colombo," Master's thesis, Institutional Repository, Univ. Moratuwa, Moratuwa, Sri Lanka, 2020. [Online]. Available: <http://dl.lib.mrt.ac.lk/handle/123/16099>
- [20] D. Prasad, A. Anand, and V. A. Sateesh, "Accident avoidance and detection on highways," in *Proc. Microelectron., Commun. Syst., Mach. Learn. Internet Thing*, vol. 2022, Jan. 2022, pp. 513–528.
- [21] S. M. Christie, R. A. Lyons, F. D. Dunstan, and S. J. Jones, "Are mobile speed cameras effective? A controlled before and after study," *Injury Prevention*, vol. 9, no. 4, pp. 302–306, Dec. 2003.
- [22] J. Tyndall, "Pedestrian deaths and large vehicles," *Econ. Transp.*, vols. 26–27, Jun. 2021, Art. no. 100219.
- [23] N. Rodrigo, "Vehicle crashes in minnesota: Spatial and temporal patterns and contributing factors to crash severity," Ph.D. dissertation, Univ. North Dakota, Grand Forks, ND, USA, 2024.
- [24] S. D. Doecke, M. R. J. Baldock, and C. N. Kloeden, "Impact speed and the risk of serious injury in vehicle crashes," *Accident Anal. Prevention*, vol. 144, Jul. 2020, Art. no. 105629.
- [25] L. Wahab and H. Jiang, "A comparative study on machine learning based algorithms for prediction of motorcycle crash severity," *PLoS ONE*, vol. 14, no. 4, Apr. 2019, Art. no. e0214966.
- [26] A. Mofolasayo, "Complete street concept, and ensuring safety of vulnerable road users," *Transp. Res. Proc.*, vol. 48, pp. 1142–1165, Jan. 2020.
- [27] C. Lin, D. Wu, H. Liu, X. Xia, and N. Bhattarai, "Factor identification and prediction for teen driver crash severity using machine learning: A case study," *Appl. Sci.*, vol. 10, no. 5, p. 1675, Mar. 2020.
- [28] M.-M. Chen and M.-C. Chen, "Modeling road accident severity with comparisons of logistic regression, decision tree and random forest," *Information*, vol. 11, no. 5, p. 270, May 2020.
- [29] J. Yuan, M. Abdel-Aty, Y. Gong, and Q. Cai, "Real-time crash risk prediction using long short-term memory recurrent neural network," *Transp. Res. Record, J. Transp. Res. Board*, vol. 2673, no. 4, pp. 314–326, Apr. 2019.
- [30] Z. Ma, G. Mei, and S. Cuomo, "An analytic framework using deep learning for prediction of traffic accident injury severity based on contributing factors," *Accident Anal. Prevention*, vol. 160, Sep. 2021, Art. no. 106322.
- [31] M. Sameen and B. Pradhan, "Severity prediction of traffic accidents with recurrent neural networks," *Appl. Sci.*, vol. 7, no. 6, p. 476, Jun. 2017.
- [32] M. Zheng, T. Li, R. Zhu, J. Chen, Z. Ma, M. Tang, Z. Cui, and Z. Wang, "Traffic Accident's severity prediction: A deep-learning approach-based CNN network," *IEEE Access*, vol. 7, pp. 39897–39910, 2019.
- [33] H. Dong and I. Kotenko, "Multi-task learning for IoT traffic classification: A comparative analysis of deep autoencoders," *Future Gener. Comput. Syst.*, vol. 158, pp. 242–254, Sep. 2024.
- [34] P. Tirat, S. Yoon, and J.-G. Lee, "MG-TAR: Multi-view graph convolutional networks for traffic accident risk prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 4, pp. 3779–3794, Apr. 2023.
- [35] G. Zou, Z. Lai, T. Wang, Z. Liu, J. Bao, C. Ma, Y. Li, and J. Fan, "Multi-task-based spatiotemporal generative inference network: A novel framework for predicting the highway traffic speed," *Expert Syst. Appl.*, vol. 237, Mar. 2024, Art. no. 121548.
- [36] H. Zhou, K. Jiang, S. He, G. Min, and J. Wu, "Distributed deep multi-agent reinforcement learning for cooperative edge caching in Internet-of-Vehicles," *IEEE Trans. Wireless Commun.*, vol. 22, no. 12, pp. 9595–9609, 2023.
- [37] D. Meng, J. Guo, H. Zhou, Y. Zhang, L. Zhao, Y. Shu, and X. Fan, "Incentive-driven partial offloading and resource allocation in vehicular edge computing networks," *IEEE Internet Things J.*, vol. 12, no. 8, pp. 11023–11035, Apr. 2025.
- [38] O. H. Kwon, W. Rhee, and Y. Yoon, "Application of classification algorithms for analysis of road safety risk factor dependencies," *Accident Anal. Prevention*, vol. 75, pp. 1–15, Feb. 2015.
- [39] S. Bramhall, H. Horn, and M. Tieu, "Qlime—A quadratic local interpretable model-agnostic explanation approach," *SMU Data Sci. Rev.*, vol. 3, no. 1, p. 4, Jan. 2020.
- [40] O. I. Aboulola, "Improving traffic accident severity prediction using MobileNet transfer learning model and SHAP XAI technique," *PLoS ONE*, vol. 19, no. 4, Apr. 2024, Art. no. e0300640.

- [41] M. Zahid, M. F. Habib, M. Ijaz, I. Ameer, I. Ullah, T. Ahmed, and Z. He, "Factors affecting injury severity in motorcycle crashes: Different age groups analysis using catboost and SHAP techniques," *Traffic Injury Prevention*, vol. 25, no. 3, pp. 472–481, Apr. 2024.
- [42] Y. Xiao and Z. Duan, "An integrated data-driven and causality crash severity model," *Math. Comput. Model. Dyn. Syst.*, vol. 31, no. 1, Dec. 2025, Art. no. 2509514.
- [43] N. Srivastava, B. N. Gohil, and S. Ray, "Road traffic accident severity prediction using causal inference and machine learning," in *Proc. 8th Int. Conf. Data Sci. Manage. Data*, Dec. 2024, pp. 292–300.
- [44] A. Shrikumar, P. Greenside, and A. Kundaje, "Learning important features through propagating activation differences," in *Proc. Int. Conf. Mach. Learn.*, vol. 2017, Jan. 2017, pp. 3145–3153.
- [45] C. Arteaga, A. Paz, and J. W. Park, "Injury severity on traffic crashes: A text mining with an interpretable machine-learning approach," *Saf. Sci.*, vol. 132, May 2020, Art. no. 104988.
- [46] J. Wang, C. Ma, S. Wang, X. Lu, and D. Li, "Risk assessment model and sensitivity analysis of ordinary arterial highways based on RSR–CRITIC–LVSSM–EFAST," *Sustainability*, vol. 14, no. 23, p. 16096, Dec. 2022.
- [47] Y. Gorishniy, I. Rubachev, and V. Khurlov, "Revisiting deep learning models for tabular data," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, Dec. 2021, pp. 18932–18943.
- [48] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.
- [49] Q. Wang, B. Li, T. Xiao, J. Zhu, C. Li, D. F. Wong, and L. S. Chao, "Learning deep transformer models for machine translation," 2019, *arXiv:1906.01787*.
- [50] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.
- [51] N. Srivastava, G. Hinton, and A. Krizhevsky, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, Jan. 2014.
- [52] M. Teichmann, M. Weber, M. Zöllner, R. Cipolla, and R. Urtasun, "Multi-Net: Real-time joint semantic reasoning for autonomous driving," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2018, pp. 1013–1020.
- [53] R. Cipolla, Y. Gal, and A. Kendall, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7482–7491.
- [54] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
- [55] S. Zhang, C. He, Z. Wan, N. Shi, B. Wang, X. Liu, and D. Hou, "Diagnosis of pulmonary tuberculosis using 3D neural network based on multi-scale attention mechanism," *Med. Biol. Eng. Comput.*, vol. 62, no. 5, pp. 1589–1600, May 2024.
- [56] X. Zhao, Z. Li, C. Zhao, R. Fu, and C. Wang, "Distraction-level recognition based on stacking ensemble learning for IVIS secondary tasks," *Expert Syst. Appl.*, vol. 244, Jun. 2024, Art. no. 122849.
- [57] Z. Chen, V. Badrinarayanan, and C. Y. Lee, "GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks," in *Proc. Int. Conf. Mach. Learn.*, Jun. 2018, pp. 794–803.
- [58] R. A. Disha and S. Waheed, "Performance analysis of machine learning models for intrusion detection system using Gini impurity-based weighted random forest (GIWRF) feature selection technique," *Cybersecurity*, vol. 5, no. 1, p. 1, Dec. 2022.
- [59] M. Madhwaran and S. N. Deepa, "Comparative analysis on hidden neurons estimation in multi layer perceptron neural networks for wind speed forecasting," *Artif. Intell. Rev.*, vol. 48, no. 4, pp. 449–471, Dec. 2017.
- [60] E. Youn, L. Koenig, M. K. Jeong, and S. H. Baek, "Support vector-based feature selection using Fisher's linear discriminant and support vector machine," *Expert Syst. Appl.*, vol. 37, no. 9, pp. 6148–6156, Sep. 2010.
- [61] S. Ahmed, M. A. Hossain, and S. K. Ray, "A study on road accident prediction and contributing factors using explainable machine learning models: Analysis and performance," *Transp. Res. Interdiscipl. Perspect.*, vol. 19, May 2023, Art. no. 100814.
- [62] P. Wei, Z. Lu, and J. Song, "Regional and parametric sensitivity analysis of sobol' Indices," *Rel. Eng. Syst. Saf.*, vol. 137, pp. 87–100, Apr. 2015.
- [63] G. Azimi, A. Rahimi, H. Asgari, and X. Jin, "Severity analysis for large truck rollover crashes using a random parameter ordered logit model," *Accident Anal. Prevention*, vol. 135, Feb. 2020, Art. no. 105355.
- [64] R. Dzinyela, N. Alnawmasi, E. Kofi Adanu, B. Dadashova, D. Lord, and F. Mannering, "A multi-year statistical analysis of driver injury severities in single-vehicle freeway crashes with and without airbags deployed," *Analytic Methods Accident Res.*, vol. 41, Mar. 2024, Art. no. 100317.
- [65] A. K. Yadav and N. R. Velaga, "Alcohol-impaired driving in rural and urban road environments: Effect on speeding behaviour and crash probabilities," *Accident Anal. Prevention*, vol. 140, Jun. 2020, Art. no. 105512.
- [66] E. K. Adanu, W. Agyemang, R. Islam, and S. Jones, "A comprehensive analysis of factors that influence interstate highway crash severity in Alabama," *J. Transp. Saf. Secur.*, vol. 14, no. 9, pp. 1552–1576, Sep. 2022.
- [67] G. Shiran, R. Imaninasab, and R. Khayamim, "Crash severity analysis of highways based on multinomial logistic regression model, decision tree techniques, and artificial neural network: A modeling comparison," *Sustainability*, vol. 13, no. 10, p. 5670, May 2021.
- [68] A. Chehri, N. Quadar, and R. Saadane, "Communication and localization techniques in VANET network for intelligent traffic system in smart cities: A review," in *Proc. 3rd KES-STIS Int. Symp. Smart Transp. Syst.*, Jan. 2020, pp. 167–177.



YAHUI WANG received the Ph.D. degree in human factors/industrial design from Northwestern Polytechnical University, in 2019. He is currently an Assistant Professor with Beijing Institute of Technology, Beijing, China. His research interests include human factors, intelligent transportation, human AI interaction, and engineering psychology.



ZHOUSHUO LIANG was born in Shandong, China, in 2001. He received the B.S. degree from Wuhan University of Technology, Wuhan, China, in 2023. He is currently pursuing the M.S. degree with the School of Medical Technology, Beijing Institute of Technology, China. His research interests include human-computer interaction, machine behavior, and human factors engineering.



YUE HE (Member, IEEE) received the B.S. degree in the Internet of Things (IoT) engineering from Zhengzhou University, Zhengzhou, China, in 2024. She is currently pursuing the master's degree in biomedical engineering with the School of Medical Technology, Beijing Institute of Technology, China. Her research interests include human-computer interaction, machine behavior, and human factors engineering.

...